

MISO DPP 2020 Cycle1 Michigan Area Study Phase 1 Report

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(Final)

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1. Executive Summary

This report presents the results of a System Impact Study (SIS) performed to evaluate the interconnection of the generators in the DPP 2020 Cycle1 Michigan Area Phase 1 (Michigan Area DPP-2020-Cycle1 Phase 1). The study was performed under the direction of MISO and reviewed by an ad hoc study group. The ad hoc study group was formed to review the study scope, methodology, models and results. The ad hoc study group consisted of representatives from the interconnection customers and the following utility companies:

- ATC
- MPower
- Consumers Energy
- PJM
- NIPSCO
- DTE
- Wolverine
- ITC

1.1. Project List

The original interconnection requests for DPP 2020 Cycle1 Michigan Area had a total of 29 generation projects: J1492, J1516, J1525, J1526, J1527, J1537, J1550, J1552, J1553, J1557, J1571, J1586, J1587, J1589, J1603, J1609, J1614, J1635, J1639, J1654, J1658, J1659, J1660, J1663, J1664, J1665, J1666, J1667 and J1798. Therefore, there are 29 projects generation projects with a combined nameplate rating of 5149 MW (ERIS) & 5174 MW (NRIS). The detailed list of Michigan Area DPP Phase 1 is shown below in Table 1, and the Phase 1 study was kicked off on February 25, 2021.

Table 1: List of DPP 2020 Cycle 1 Michigan Area Phase 1 Projects

Project #	Fuel	Summer /Winter MW	Transmission Owner	POI Name	County	State	Service Type
J1492	Battery Storage	150	METC	Gaines 138kV	Kent	MI	NRIS
J1516	Solar	100	ITC	Fitz 120kV	St. Clair	MI	NRIS
J1525	Solar	200	ITC	Grassmere 345kV	Huron	MI	NRIS
J1526	Solar	200	ITC	Greenwood 345kV	St. Clair	MI	NRIS
J1527	Solar	200	METC	Kenowa 345kV	Ottawa	MI	NRIS
J1537	Solar	150	METC	Thetford #1 - Dort 138kV T-Line	Genesee	MI	NRIS
J1550	Solar	200	METC	Batavia - Tap Kinderhook distribution station, 138kV Transmission Line	Branch	MI	NRIS
J1552	Solar	100	METC	Blackstone to Marshall 138kV (J1310 POI)	Jackson	MI	NRIS
J1553	Solar	150	METC	Bingham West 138kV	Clinton	MI	NRIS
J1557	Wind	293	ITC	Greenwood to Banner 345 kV Line	Sanilac	MI	NRIS
J1571	Solar	200	METC	Morocco to Beecher 138kV (Ckt 1)	Lenawee	MI	NRIS
J1586	Hybrid	499	METC	Argenta to Tompkins 345kV line	Calhoun	MI	NRIS
J1587	Solar	100	METC	Du Pont to Cobb 138 KV (J1375 POI)	Muskegon	MI	NRIS
J1589	Solar	150	METC	Hubbardston - Tech Drive 138kV	Clinton	MI	NRIS
J1603	Solar	200	METC	Vantyle 138kV	Otsego	MI	NRIS
J1609	Solar	125	METC	Batavia-Jaguar 138 kV	St. Joseph	MI	NRIS

Project #	Fuel	Summer /Winter MW	Transmission Owner	POI Name	County	State	Service Type
J1614	Solar	250	METC	White Road - Four Mile 138kV line	Muskegon	MI	NRIS
J1635	Battery Storage	100	METC	138kV Michigan Ave to Newton	Branch	MI	NRIS
J1639	Battery Storage	100	METC	345kV Allen Junction to Morocco	Lenawee	MI	NRIS
J1654	Solar	200	METC	Goss-Thetford 345kV Line	Genesee	MI	NRIS
J1658	Solar	180	METC	Coldwater to Moore Road 138kV Line	Hillsdale	MI	NRIS
J1659	Battery Storage	100	ITC	120 kV Dorset	Washtenaw	MI	NRIS
J1660	Battery Storage	50	METC	Goss-Thetford 345kV Line	Genesee	MI	NRIS
J1663	Solar	200	METC	Argenta to Tallmadge 345kV	Ottawa	MI	NRIS
J1664	Solar	200	ITC	Carrigan 120kV Substation	St. Clair	MI	NRIS
J1665	Battery Storage	50	ITC	Carrigan 120kV Substation	St. Clair	MI	NRIS
J1666	Wind	300	METC	Oneida - Majestic 345 kV	Ingham	MI	NRIS
J1667	Wind	200	METC	Bard Rd to Clare Junction 138 kV	Isabella,Midland	MI	NRIS
J1798	Gas	25	ITC	Waterman 120kV	Wayne	MI	ERIS

1.2. Total Network Upgrades

The cost allocation of Network Upgrades for the projects in the DPP 2020 Cycle1 Michigan Phase 1 reflects responsibilities for mitigating system impacts. The total cost of network upgrades is listed in Table 2 below. The costs for Network Upgrades are planning-level estimates and subject to revision in the facility studies. Details pertaining to the cost allocation has been detailed in [Appendix A – Cost Allocation Summary \(CEII\)](#)

Table 2: Total Cost of Network Upgrades for DPP 2020 Cycle1 Michigan Phase 1 Projects

Project	DPP Phase II									Payment in DPP Phase I		Payment in DPP Phase II		Interconnection Facilities - Self Fund
	ERIS Network Upgrades (\$)				NRIS Network Upgrades (\$)	Interconnection Facilities (\$)		Shared Network Upgrades (\$)	Total Cost of NU (\$)	M2 (\$)	M3 =10% of NU-M2 (\$)	20% of Total Cost of NU (\$)	M4= 20% of NU-M2-M3 (\$)	
	Thermal	Stability	Short Circuit	Affected System	Deliverability	TO Network Upgrades	TO-Owned Direct Assigned							
a	b	c	d	e	f	g	h	i	j = b+c+d+f+g+i					
J1492					\$2,441,874.61	\$2,569,148.00	\$500,000.00		\$5,011,022.61	\$600,000.00	\$0.00			No
J1516					\$2,214,950.76	\$1,039,319.00	\$500,000.00		\$3,254,269.76	\$400,000.00	\$0.00			No
J1525					\$4,053,357.29	\$5,467,854.00	\$500,000.00		\$9,521,211.29	\$800,000.00	\$152,121.13			No
J1526					\$4,370,151.31	\$1,858,285.00	\$500,000.00		\$6,228,436.31	\$800,000.00	\$0.00			No
J1527					\$1,955,054.35	\$3,244,798.00	\$500,000.00		\$5,199,852.35	\$800,000.00	\$0.00			No
J1537					\$4,146,464.52	\$5,464,166.00	\$500,000.00		\$9,610,630.52	\$600,000.00	\$361,063.05			No
J1550	\$6,119,830.70				\$15,238,370.17	\$8,833,980.00	\$500,000.00		\$30,192,180.87	\$800,000.00	\$2,219,218.09			No
J1552	\$11,025,476.15				\$13,617,583.68	\$0.00	\$0.00		\$24,643,059.83	\$400,000.00	\$2,064,305.98			N/A
J1553					\$4,717,522.04	\$5,462,290.00	\$500,000.00		\$10,179,812.04	\$600,000.00	\$417,981.20			No
J1557					\$6,366,675.25	\$9,792,526.00	\$500,000.00		\$16,159,201.25	\$1,172,000.00	\$443,920.12			No
J1571	\$15,557,638.34				\$35,663,884.29	\$5,464,166.00	\$500,000.00		\$56,685,688.63	\$800,000.00	\$4,868,568.86			No
J1586	\$79,272,622.55				\$8,139,101.09	\$10,903,183.00	\$500,000.00		\$98,314,906.64	\$1,996,000.00	\$7,835,490.66			No
J1587					\$986,882.95	\$116,911.00	\$0.00		\$1,103,793.95	\$400,000.00	\$0.00			No
J1589					\$4,340,029.80	\$11,768,417.00	\$500,000.00		\$16,108,446.80	\$600,000.00	\$1,010,844.68			No
J1603					\$2,809,354.28	\$1,039,319.00	\$500,000.00		\$3,848,673.28	\$800,000.00	\$0.00			No
J1609	\$3,262,035.25				\$5,623,047.83	\$6,404,495.00	\$500,000.00		\$15,289,578.08	\$500,000.00	\$1,028,957.81			No
J1614	\$2,100,000.00				\$2,413,511.36	\$7,842,573.00	\$500,000.00		\$12,356,084.36	\$1,000,000.00	\$235,608.44			No
J1635	\$4,918,287.98				\$11,664,545.25	\$5,704,135.00	\$500,000.00		\$22,286,968.22	\$400,000.00	\$1,828,696.82			No
J1639					\$1,572,630.68	\$12,902,345.00	\$500,000.00		\$14,474,975.68	\$400,000.00	\$1,047,497.57			No
J1654	\$9,614,647.79				\$7,870,719.64	\$13,018,491.00	\$1,000,000.00		\$30,503,858.44	\$800,000.00	\$2,250,385.84			No
J1658	\$18,494,177.61				\$28,279,926.48	\$11,157,938.00	\$500,000.00		\$57,932,042.09	\$720,000.00	\$5,073,204.21			No
J1659					\$2,812,771.92	\$487,882.00	\$742,931.00		\$3,300,653.92	\$400,000.00	\$0.00			No
J1660	\$2,385,352.21				\$1,967,085.51	\$0.00	\$0.00		\$4,352,437.72	\$200,000.00	\$235,243.77			N/A
J1663					\$2,271,287.87	\$10,903,183.00	\$500,000.00		\$13,174,470.87	\$800,000.00	\$517,447.09			No
J1664	\$21,786,525.39				\$4,401,667.03	\$922,408.00	\$500,000.00		\$27,110,600.41	\$800,000.00	\$1,911,060.04			No
J1665	\$5,413,474.61				\$1,100,374.12	\$922,408.00	\$500,000.00		\$7,436,256.73	\$200,000.00	\$543,625.67			No
J1666	\$1,781,931.86				\$5,716,553.43	\$10,786,272.00	\$500,000.00		\$18,284,757.29	\$1,200,000.00	\$628,475.73			No
J1667	\$23,300,000.00				\$3,116,165.15	\$10,155,270.00	\$500,000.00		\$36,571,435.15	\$800,000.00	\$2,857,143.52			No
J1798	\$100,000.00				\$0.00	\$0.00	\$0.00		\$100,000.00	\$0.00	\$10,000.00			N/A
Total	\$205,132,000.43	\$0.00	\$0.00	\$0.00	\$189,871,542.66	\$164,231,762.00	\$13,242,931.00	\$0.00	\$559,235,305.09	\$19,788,000.00	\$37,540,860.29			

Analyses performed demonstrate the following transmission facilities are required to reliably interconnect this group of generators to the transmission system. Energy Resource Interconnection Service (ERIS) Network Upgrades and Network Resource Interconnection Service (NRIS) Network Upgrades are shown in Table 3. No Shared Network Upgrades was identified for this cycle.

Table 3: ERIS & NRIS Upgrades (Planning level cost estimates)

Element Name	Network Upgrade	TO	GI projects requiring upgrade for ERIS	GI projects requiring upgrade for NRIS	Cost of solution (\$)	A10/ISD	MTEP cost of solution (\$)	Notes
White Lake - Dupont 138kV	Covered under MTEP22 #22021 - Target A	218	J1587		\$0.00	12/31/2025	\$39,002,014.00	If this MTEP project is not approved then cost will be allocated to the IC
J1375 - Dupont 138kV	Covered under MTEP22 #22021 - Target A	218	J1587		\$0.00	12/31/2025	\$39,002,014.00	If this MTEP project is not approved then cost will be allocated to the IC
Chopin - White Lake 138kV	Covered under MTEP22 #22021 - Target A	218	J1587		\$0.00	12/31/2025	\$39,002,014.00	If this MTEP project is not approved then cost will be allocated to the IC
Goss Transformer 1	Install a second Transformer at Goss 345/138kV station	218	J1654,J1660		\$12,000,000.00			
Tompkins Transformer 2	Loop in the BattleCreek - J1307 line into J1586	218	J1586		\$6,200,000.00			
Tompkins Transformer 1	Loop in the BattleCreek - J1307 line into J1586	218	J1586		\$6,200,000.00			
Oneida Transformer 2	Loop in the BattleCreek - J1307 line into J1588	218	J1666		\$6,200,002.00			
Bingham - Cornell (Bluewater Junction - Bingham) 138kV	MTEP21# 20015 Raises limit to 288 MVA ISD 6/1/2025	218	J1553		\$0.00	6/1/2025	\$31,000,000.00	If this MTEP project is not approved then cost will be allocated to the IC
Bingham - Cornell (Bluewater Junction - Cornell) 138kV	MTEP21# 20015 Raises limit to 288 MVA ISD 6/1/2025	218	J1553		\$0.00	6/1/2025	\$31,000,000.00	If this MTEP project is not approved then cost will be allocated to the IC
Bingham - Cornell (Bluewater Junction - Bingham) 138kV	MTEP21# 20015 Raises limit to 288 MVA ISD 6/1/2025	218	J1589		\$0.00	6/1/2025	\$31,000,000.00	If this MTEP project is not approved then cost will be allocated to the IC

Element Name	Network Upgrade	TO	GI projects requiring upgrade for ERI5	GI projects requiring upgrade for NRIS	Cost of solution (\$)	A10/ISD	MTEP cost of solution (\$)	Notes
Bingham - Cornell (Bluewater Junction - Cornell) 138kV	MTEP21# 20015 Raises limit to 288 MVA ISD 6/1/2025	218	J1589		\$0.00	6/1/2025	\$31,000,000.00	If this MTEP project is not approved then cost will be allocated to the IC
Bingham - J1589 POI (Bingham - TechDrive Junciton) 138kV	MTEP22# 15910 raises limit to 288 MVA ISD 2025	218	J1589		\$0.00	6/1/2025	\$22,100,000.00	If this MTEP project is not approved then cost will be allocated to the IC
Bingham - J1589 POI (J1589 POI - TechDrive Junciton) 138kV	MTEP22# 15910 raises limit to 288 MVA ISD 2025	218	J1589		\$0.00	6/1/2025	\$22,100,000.00	If this MTEP project is not approved then cost will be allocated to the IC
Beecher - Knowles (Beecher - Dowling Junction) 138kV	Raise sag limit to 189 MVA (3 POIs)	218	J1658		\$900,000.00			
Coldwater - J1658 POI (Coldwater - Ash Road Junction) 138kV	Raise Sag Limit to 198 MVA SE (6 POIs)	218	J1635		\$1,800,000.00			
Beecher - Knowles (Knowles - Dowling Junction) 138kv	Raise sag limit to 244 MVA (14 POIs)	218	J1658		\$4,200,000.00			
Majestic - Wayne 345kV	Raise Sag Limit to 984 MVA SE (1 POI)	219	J1666		\$300,000.00			
Higgins - Bard Road (Bard Road - Backus J) 138kV	Raise Sag to 117 MVA SE (1 POIs)	218	J1667		\$300,000.00			
Chopin - J1614 POI (Chopin - White Road Junction) 138kV	Raise Sag to 273 MVA SE (1 POI)	218	J1614		\$300,000.00			
Chopin - J1614 POI (J1614 POI - White Road Junction) 138kV	Raise Sag to 277 MVA SE (3 POIs)	218	J1614		\$900,000.00			
Meyer - J1614 POI 138KV	Raise the sag limit to 250 MVA SE (3 POIs)	218	J1614		\$900,000.00			
Pixley - Morocco 138kV	Rebuild 11.37mi of 954ACSR to 1431 ACSR	801/218	J1571,J1658		\$20,700,000.00			
Redstone - Tittabawassee 138kV	Rebuild 12.6 miles of 336 ACSR to 1431 ACSR	218	J1667		\$23,000,000.00			
J1320 POI - Coldwater 138kV	Rebuild 2 miles of 477ACSR and 366 ACSS with 954 ACSR	901/218	J1550,J1609		\$3,600,000.00			
Majestic - Tompkins 345kV	Rebuild 26 miles 2156 ACSR with 2-954 ACSR and raise Sag limit to 1641 MVA SE	218/219	J1586,J1552,J1550,J1658,J1635,J1609		\$73,632,000.00			
J1658 POI - Moore Road 138kV	Rebuild 4.4 miles with 1431 ACSR, moore terminal work & moore bkr 33b7	218	J1658,J1550,J1635,J1609		\$8,000,000.00			
Bunce - Menlo 120kV	Rebuild 6.3 miles of 266.8ACSR with 954 ACSR	219	J1664,J1665		\$11,500,000.00			
Carrigan - Menlo 120kV	Rebuild 8.6 miles of 477 ACSR and 266.8ACSR with 954 ACSR	219	J1664,J1665		\$15,700,000.00			
J1310 POI - Blackstone (Blackstone - Maines Road Junction) 138kV	Reconductor 2.8mi of 954 ACSR to 1431 ACSR & upgrade BLKSN 377 terminal	218	J1552		\$5,100,000.00			

Element Name	Network Upgrade	TO	GI projects requiring upgrade for ERI5	GI projects requiring upgrade for NRIS	Cost of solution (\$)	A10/ISD	MTEP cost of solution (\$)	Notes
J1310 POI - Blackstone (J1310 POI - Maines Road Junction) 138kV	Reconductor 2.8mi of 954 ACSR to 1431 ACSR & upgrade BLKSN 377 terminal	218	J1552		\$5,100,000.00			
Tompkins - J1586 POI 345kV	Reconductor 20 miles with 1431 ACSR	218	J1586		\$19,500,000.00			
Dig - Navarre - Waterman (Dig - Dig Tap) 230 kV	Replace terminal equipment new limit will be 598MVA SN, 719MVA SE	219	J1798		\$100,000.00			
Beecher 500 Breaker	Upgrade Relay Thermal limit to at least 340 MVA	218	J1571,J1658		\$100,000.00			
Beecher - Morocco (Beecher - MPG Junction) #2 138kV	MTEP15 8118 updated rating to be 288MVA SN, 332 MVA SE	218	J1571,J1658		N/A			
J1639 POI - Morocco 345kV	Build 8.3 miles for a second J1639 - Morocco 345kV line.	218		J1492, J1516, J1525, J1526, J1527, J1537, J1550, J1552, J1553, J1557, J1571, J1586, J1587, J1589, J1603, J1609, J1614, J1635, J1654, J1658, J1659, J1660, J1663, J1664, J1665, J1666, J1667	\$28,000,000.00			
J1639 POI - Allen Junction 345kV #1&2	Build Second line from Allen - J1639. FE to upgrade limits to at least 1900 MVA (cost does not include FE costs)	218/202		J1492, J1516, J1525, J1526, J1527, J1537, J1550, J1552, J1553, J1557, J1571, J1586, J1587, J1589, J1603, J1609, J1614, J1635, J1639, J1654, J1658, J1659, J1660, J1663, J1664, J1665, J1666, J1667	\$3,500,000.00			
Gallagher - Twining (Twining - Withey Lake) 138kV	Covered in MTEP21 #20049 (MTEP 21 Target A)	218		J1667	\$0.00	6/1/2025	\$971,000.00	If this MTEP project is not approved then cost will be allocated to the IC
Beecher 500 Breaker	Increase Relay Limit to 390 MVA	218		J1550, J1609, J1635	\$100,000.00			
Gaines - J1292 POI (Gaines - Caledonia Junction) 138kV	Increase Sag limit to 195 MVA SE	218		J1553, J1589	\$1,000,000.00			
Morocco Transformer #2	Install 2nd Transformer at Morocco	218		J1550, J1552, J1571, J1586, J1609, J1635, J1658, J1659	\$39,002,014.00			
Bullock - Shale (Gleaner Junciton - Hackett Junction) 138kV	MTEP 20018 raises limits to 288 SN, 332 SE (MTEP21 Target A)	218		J1667	\$0.00	6/1/2025	\$33,000,000.00	If this MTEP project is not approved then cost will be allocated to the IC
Bullock - Shale (Hackett Junction - Laundra Junction) 138kV	MTEP 20018 raises limits to 288 SN, 332 SE (MTEP21 Target A)	218		J1667	\$0.00	6/1/2025	\$33,000,000.00	If this MTEP project is not approved then cost will be allocated to the IC
Bullock - Shale (Bullock - Gleaner Junction) 138kV	MTEP 20018 raises limits to 288 SN, 332 SE (MTEP21 Target A)	218		J1667	\$0.00	6/1/2025	\$33,000,000.00	If this MTEP project is not approved then cost will be allocated to the IC

Element Name	Network Upgrade	TO	GI projects requiring upgrade for ERI	GI projects requiring upgrade for NRIS	Cost of solution (\$)	A10/ISD	MTEP cost of solution (\$)	Notes
Bullock - Shale (Shale - Laundra Junction) 138kV	MTEP 20018 raises limits to 288 SN, 332 SE (MTEP21 Target A)	218		J1667	\$0.00	6/1/2025	\$33,000,000.00	If this MTEP project is not approved then cost will be allocated to the IC
Bullock - Salt River (Bullock - Edenville Junction) #1 138kV	MTEP 20019 raises limit to (288 SN, 332 SE) (MTEP21 Target A)	218		J1667	\$0.00	6/1/2025	\$21,100,000.00	If this MTEP project is not approved then cost will be allocated to the IC
Bullock - Salt River #2 138kV	MTEP 20021 Raises limit to 288 SN, 332 SE (MTEP21 Target A)	218		J1667	\$0.00	12/31/2024	\$29,310,000.00	If this MTEP project is not approved then cost will be allocated to the IC
Argenta - J1586 POI 345kV	Raise sag limit to 1005 MVA SE (1 POI)	218		J1663	\$500,000.00			
Coldwater - J1658 POI (Coldwater - Ash Road Junction) 138kV	Raise sag limit to 203 MVA SE (1 POI)	218		J1550, J1609	\$300,000.00			
Beecher - Knowles (Beecher - Dowling Junction) 138kV	Raise sag limit to 225 MVA SE (2 POIs)	218		J1550, J1609, J1635	\$600,000.00			
Mud Lake - Verona #1 138kV	Rebuild (4.4 mi) Verona-MudLake #1 from 795ACSR to 1431 ACSR (372 SN, 431 SE)	218		J1552, J1571	\$7,100,000.00			This NU is proposed as part of DPP2019 phase 2 ERI mitigations. The cost is attributed to J1430, J1310, and J1248. We'd update the 954 ACSR conductor with 1431 ACSR but the cost estimate for DPP2019 wouldn't change.
Leoni - Beecher (Beecher - Cement City Junction) 138kV	Rebuild 0.2 miles of 477 ACSR with 954 ACSR. Raise sag of 336 ACSRx2 to 257 SE (227 SN, 257 SE) (8 POIs)	218		J1552	\$2,400,000.00			
Leoni - Beecher (Cement City Junciton - Seamless Junction) 138kV	Rebuild 0.5 miles of 477 with 954 ACSR, and raise the sag of the 4/0 CUX2 to at least 227 SN, 292 MVA SE (25 POIs)	218		J1552	\$1,200,000.00			
Beecher - Morocco (Beecher - MPG Junction) #2 138kV	Rebuild 12 miles of 954 aCSR to 1431 ACSR (372 SN, 432 SE)	218		J1550, J1552, J1609, J1635	\$19,300,000.00			
Monroe - Lallendorf 345kV	Rebuild 16.8 miles of 2-954 ACSR to 2-1431ACSR. Will require work from FE (SN 1861, SE 2157)	202/219		J1492, J1516, J1525, J1526, J1527, J1537, J1550, J1552, J1553, J1557, J1571, J1586, J1587, J1589, J1603, J1609, J1614, J1635, J1639, J1654, J1658, J1659, J1660, J1663, J1664, J1665, J1666, J1667	\$47,600,000.00			
J1401 POI - Wagner 138kV	Rebuild 2.11 mi of 266.8 with 954 ACSR and upgrade terminal equipment (288 SN, 332 SE)	218		J1550, J1609	\$3,900,000.00	12/31/2021		This NU is covered under MTEP #11884
Cornell - Tihart (Cornell - Bennington Junction) 138kV	Rebuild 2.33 miles of 477 ACSR to 954 ACSR and remediate sag on 795ACSR (236 SN, 236 SE) (5 POIs)	218		J1553, J1589, J1654, J1660	\$5,700,000.00			
Leoni - Beecher (Leoni - Seamless Junction) 138kV	Rebuild 2.7 miles of 4/0Cu and 795ACSR to 1431 ACSR (372 MVA SN, 416 MVA SE) and Terminal upgrades	218		J1552	\$5,000,000.00			
Cornell - J1297 POI (Cornell - North Corunna Junction) 138kV	Rebuild 3.037 miles of 477 ACSR to 954 ACSR (288SN, 332 SE)	218		J1537, J1654, J1660	\$5,000,000.00			
Beecher - Morocco (Morocco - MPG Junction) #2 138kV	Rebuild 3.5 miles of 954ACSR with 1431 ACSR (372 SN, 432 SE)	218		J1550, J1552, J1571, J1609, J1635, J1658	\$5,600,000.00			

Element Name	Network Upgrade	TO	GI projects requiring upgrade for ERIIS	GI projects requiring upgrade for NRIS	Cost of solution (\$)	A10/ISD	MTEP cost of solution (\$)	Notes
J1320 POI - Batavia 138kV	Rebuild 6.3 miles (366 ACSR) to 954 ACSR (288 SN, 332 SE)	901/218		J1635, J1658	\$11,500,000.00			
Verona - Wagner (Wagner - Alder Creek Junction) 138kV	Rebuild 7 miles of 954 ACSR to 1431 ACSR (372 SN 432 SE)	218		J1550, J1609, J1635, J1658	\$11,256,000.00			
Verona - Wagner (Alder Creek Junction - Barnum Creek Junction) 138kV	Rebuld 6.7 mi of 954 ACSR with 1431 ACSR and replace switch at Alder Creek (372 SN, 431 SE)	218		J1550, J1609, J1635, J1658	\$10,800,000.00			
J1571 POI - Beecher 138kV	Replace terminal equipment at Beecher (288 MVA SN, 332 MVA SE)	218		J1571	\$100,000.00			
256147 18GAINES 138 258073 CALEDON J 138 1	Raise sag to 133 SN, 133 SE	218		J1492	\$1,000,000.00			
264664 19STEPH 345 264752 19CANIF 345 1	Fixed with Caniff Phase Shifter	219		J1516, J1525, J1526, J1557, J1664, J1665	N/A			
256155 18GOSS 138 40893 J1297 POI 138 1	MTEP 15798 raises limits to 264 SN, 282 SE (MTEP 19 Target A)	801/218		J1537, J1654, J1660	N/A			
40893 J1297 POI 138 256115 18CORNELLB 138 1	MTEP 15798 raises limits to 264 SN, 282 SE (MTEP 19 Target A)	801/218		J1537, J1654, J1660	N/A			
256069 BENNINGTN 138 256520 18SNCLKJ 138 1	MTEP 15852 raises SE to 234MVA (MTEP 19 Target A)	218		J1553, J1589, J1654, J1660	N/A			
256520 18SNCLKJ 138 256283 18VANATJ 138 1	MTEP 15852 raises SE to 234MVA (MTEP 19 Target A)	218		J1553, J1589, J1654, J1660	N/A			

Note:

- 1) Details pertaining to upgrades, costs, and the execution plan for interconnection of the generating facility at the POI will be documented in the Facility Study for Interconnecting Generator.
- 2) Facilities that have been included as base case assumptions and the level of interconnection service that would be conditional upon these facilities being in service will be documented in the GIA (Generator Interconnection Agreement) for each respective GI request successfully achieving GIA execution.
- 3)
- 4) Analysis performed shows there are no study projects for Shared Network Upgrade cost allocation.
- 5) A detailed summary of the estimated cost of Network Upgrades on a per project basis can be found in Appendix B – Network Upgrade Table Per Project (CEII).

2. FERC Order 827 Compliance Review

The Final Rule of FERC Order 827 “Reactive Power Requirements for Non-Synchronous Generation”, which was issued June 16, 2016, stated that “Under this Final Rule, newly interconnecting non-synchronous generators that have not yet executed a Facilities Study Agreement as of the effective date of this Final Rule will be required to provide dynamic reactive power within the range of 0.95 leading to 0.95 lagging at the high-side of the generator substation.” As such, this Final Rule applies to J1192, J1194, J1196, J1203, J1206, J1207, J1210, J1211, J1224, J1226, J1248, J1255, J1262, J1292, J1297, J1307, J1310, J1319, J1320, J1322, J1327, J1329, J1331, J1350, J1370, J1375, J1379, J1389, J1394, J1399, J1401, J1403, J1430, J1449 and J1472 projects included in the DPP 2020 Cycle1 Michigan study cycle.

In this study, the power factor at the high-side of the generator substation for each inverter based project was calculated and reviewed. The study method is to set Qgen of each study project at its Qmax, solve the case, then record the P and Q injection on the high side of the generator substation to calculate the lagging power factor (injecting VAR to the system). The same process is then repeated by setting Qgen at Qmin to calculate the leading power factor (absorbing VAR from the system).

The results show that all projects meet the requirement to maintain 0.95 leading power factor, however, four projects do not meet the requirement to provide reactive power capability corresponding to 0.95 lagging power factor, as highlighted in red below in Table 4. Additional reactive support will be needed for these projects to meet the FERC requirement on reactive power capability prior to the completion of their GIA.

Table 4: FERC Order 827 Review Results and Requested MW @ POI

Project	Pmax (MW)	Reactive Power Capability (MVar)	Proposed VAR Compensation	VAR Injection			VAR Absorption			Meet FERC Order 827 Requirement?	Add'l VAR Needed (MVar)	Requested MW	POI MW	Tolerance (1%)	POI MW met?
				P (MW)	Q (MVar)	Lagging p.f (pu)	P (MW)	Q (MVar)	Leading p.f. (pu)						
J1492	158.19	172.00	#N/A	149.95	116.40	0.79	149.92	-77.03	0.89	Yes	0.00	150	149.95	1.5	Yes
J1516	104.00	33.30	#N/A	100.14	16.11	0.99	100.05	-56.30	0.87	No	16.80	100	100.14	1	Yes
J1525	207.76	67.18	#N/A	200.50	22.43	0.99	200.02	-67.20	0.95	No	43.50	200	200.5	2	Yes
J1526	207.76	67.15	#N/A	200.38	22.07	0.99	199.96	-69.10	0.95	No	43.80	200	200.38	2	Yes
J1527	207.76	67.35	#N/A	200.85	21.32	0.99	200.29	-69.27	0.95	No	44.70	200	200.85	2	Yes
J1537	152.44	73.83	6 X 5	149.82	76.65	0.89	149.54	-54.40	0.94	Yes	0.00	150	149.82	1.5	Yes
J1550	203.29	104.00	7 X 5	198.99	101.07	0.89	192.30	204.80	0.68	Yes	0.00	200	198.99	2	Yes
J1552	121.60	33.53	4 X 4	100.98	47.77	0.90	100.50	40.60	0.93	Yes	0.00	100	100.98	1	Yes
J1553	172.20	78.13	#N/A	150.43	50.35	0.95	149.97	-54.60	0.94	Yes	0.00	150	150.43	1.5	Yes
J1557	302.40	19.57	#N/A	296.20	34.10	0.99	292.30	160.60	0.88	No	63.30	293	296.2	2.93	Yes
J1571	230.00	67.05	#N/A	200.60	22.29	0.99	199.98	-70.63	0.94	No	43.60	200	200.6	2	Yes
J1586	1021.37	509.30	#N/A	499.06	201.92	0.93	492.78	308.39	0.85	Yes	0.00	499	499.06	4.99	Yes
J1587	121.60	33.53	4 X 5.25	100.90	41.50	0.92	100.60	-56.50	0.87	Yes	0.00	100	100.9	1	Yes
J1589	150.00	78.25	#N/A	148.16	47.41	0.95	147.72	-57.95	0.93	No	1.30	150	148.16	1.5	No
J1603	229.60	67.05	#N/A	200.68	23.61	0.99	200.30	-65.86	0.95	No	42.40	200	200.68	2	Yes
J1609	152.00	41.41	8 X 6	125.15	55.83	0.91	124.99	64.10	0.89	Yes	0.00	125	125.15	1.25	Yes
J1614	250.00	82.17	9 X 13.5	248.19	120.56	0.90	247.94	121.52	0.90	Yes	0.00	250	248.19	2.5	Yes
J1635	104.94	49.40	#N/A	100.58	31.47	0.95	99.75	126.55	0.62	No	1.60	100	100.58	1	Yes

Project	Pmax (MW)	Reactive Power Capability (MVar)	Proposed VAR Compensation	VAR Injection			VAR Absorption			Meet FERC Order 827 Requirement?	Add'l VAR Needed (MVar)	Requested MW	POI MW	Tolerance (1%)	POI MW met?
				P (MW)	Q (MVar)	Lagging p.f (pu)	P (MW)	Q (MVar)	Leading p.f. (pu)						
J1639	104.94	49.40	#N/A	100.55	31.03	0.96	99.89	-76.06	0.80	No	2.00	100	100.55	1	Yes
J1654	203.32	66.83	#N/A	199.34	17.86	1.00	198.63	-74.85	0.94	No	47.70	200	199.34	2	Yes
J1658	185.38	59.82	#N/A	178.62	16.69	1.00	177.77	-69.47	0.93	No	42.00	180	178.62	1.8	Yes
J1659	104.94	49.40	#N/A	100.57	31.30	0.95	99.75	-77.83	0.79	No	1.80	100	100.57	1	Yes
J1660	53.82	16.58	#N/A	50.03	8.18	0.99	49.91	-27.56	0.88	No	8.30	50	50.03	0.5	Yes
J1663	229.60	67.05	#N/A	200.62	19.90	1.00	200.05	-72.77	0.94	No	46.00	200	200.62	2	Yes
J1664	206.31	66.39	#N/A	198.64	45.90	0.97	194.95	111.79	0.87	No	19.40	200	198.64	2	Yes
J1665	53.82	16.50	#N/A	50.00	15.10	0.96	49.99	-18.75	0.94	No	1.30	50	50	0.5	Yes
J1666	302.40	19.72	#N/A	294.60	11.20	1.00	287.30	258.60	0.74	No	85.60	300	294.6	3	No
J1667	201.60	13.19	#N/A	197.30	79.20	0.93	193.10	-92.80	0.90	Yes	0.00	200	197.3	2	No
J1798	34.00	18.00	#N/A	64.97	-22.09	0.95	64.36	-26.06	0.93	Yes	0.00	25	64.97	0.25	Yes

3. Model Development and Study Assumptions

3.1. Base Case Models

The origin of the DPP 2020 Cycle1 Michigan models is the MTEP 19 models with the Bench Cases including all pre-queued projects and associated network upgrades, while the Study Cases contain all of the interconnection requests in DPP 2020 Cycle1 Michigan Phase 1, in addition to all the facilities in the Bench Cases. Table 5 provides the list of different scenario's study for DPP 2020 Cycle1 Michigan.

- Bench Cases
 - DPP20-2025SH90-Phasel-Bench_MI_rev1
 - DPP20-2025SUM-Phasel-Bench_MI_rev1
- Study Cases
 - DPP20-2025SH90-Phasel-Study_MI_rev1
 - DPP20-2025SUM-Phasel-Study_MI_rev1
 - DPP20-2025SH90-Phasel-Study_Charging_MI_rev1

Table 5: DPP 2020 Cycle1 Michigan-Scenarios

Case Category	Case Name	Case Season	Flow on ITCT-IESO interface	Flow on METC-ATC DC line	Status of Ludington	Load Level
1	DPP20-2025SUM-Phasel-Study_MI_rev1	Summer Peak 2025	Zero Flow	5 MW ATC to METC	Generating (6 Gen)	Peak
2	DPP20-2025SH90-Phasel-Study_MI_rev1 DPP20-2025SH90-Phasel-Study_Charging_MI_rev1	Summer Shoulder 2025	Zero Flow	5 MW ATC to METC	Generating (5 Gen)	MTEP 18 Base SH (77-78%)

3.2. Monitored Elements

Under NERC category P0 conditions (system intact) branches were monitored for loading above the normal rating (PSS®E Rating A), and for NERC category P1-P7 conditions branches were monitored for emergency

rating (PSS®E Rating B). Voltage limits were specified for system intact and contingent conditions as per applicable Transmission Owner Planning Criteria.

3.3. Contingencies

The following contingencies were considered in the steady state analysis:

- 1) NERC Category P0 (system intact -- no contingencies)
- 2) NERC Category P1 contingencies
 - a. Single element outages, at buses with a nominal voltage of 68 kV and above
 - b. Multiple element NERC Category P1 contingencies
- 3) NERC Category P2-P7 contingencies
- 4) For all the contingencies and post-disturbance analyses, cases were solved with transformer tap adjustment enabled, area interchange adjustment disabled, phase shifter adjustment disabled (fixed) and switched shunt adjustment enabled.

3.4. Study Methodology

Non-linear (AC) contingency analysis was performed on the benchmark and study cases, and the incremental impact of the DPP 2020 Cycle1 Michigan generating facilities was evaluated by comparing the steady state performance of the transmission system in the Bench and Study Cases. Analyses used PSS®E version 34.6.1 and TARA version 2001_2_0_0.

3.5. Performance Criteria

A branch is considered a thermal constraint if the following conditions are met:

- 1) The generator has a larger than twenty percent (20%) sensitivity factor on the overloaded facilities under post-contingent condition (see NERC TPL) or five percent (5%) sensitivity factor under system-intact condition, or
- 2) The overloaded facility or the overload-causing contingency is at generator's outlet, or
- 3) The megawatt impact due to the generator is greater than or equal to twenty percent (20%) of the applicable rating (normal or emergency) of the overloaded facility, or
- 4) For any other constrained facility, where none of the Study Generators meet one of the above criteria, however, the cumulative MW impact of the group of study generators is greater than twenty percent (20%) of the rating of the facility, then only those study generators whose individual MW impact is greater than five percent (5%) of the rating of the facility and has DF greater than five percent (5%) will be responsible for mitigating the cumulative MW impact constraint, or
- 5) Impacts on Affected Systems would be classified as Injection constraints based on the Affected Systems' criteria, or
- 6) Any other applicable Transmission Owner FERC filed Local Planning Criteria are met.

A bus is considered a voltage constraint if both of the following conditions are met:

- 1) The bus voltage is outside of the applicable normal or emergency limits for the post change case, and
- 2) The change in bus voltage is greater than 0.01 per unit

All generators must mitigate thermal injection constraints and voltage constraints in order to obtain any type of Interconnection Service. Further, all generators requesting Network Resource Interconnection Service (NRIS) must mitigate constraints found by using the Deliverability algorithm, to meet the system performance criteria for NERC category P1 events, if DFAX due to the study generator is equal to or greater than 5%.

4. Thermal Analysis

The thermal analysis results for 2025 Summer Peak and Summer Shoulder scenarios show generator projects J1550, J1552, J1571, J1586, J1609, J1614, J1635, J1654, J1658, J1660, J1664, J1665, J1666, J1667, J1798 causing constraints. The details pertaining to the thermal analysis can be found in Appendix A – Cost Allocation Summary (CEII)

5. Voltage Analysis

The voltage analysis results for 2025 Summer Peak and Summer Shoulder scenarios show that the study generators do not cause any voltage violations. The details pertaining to the voltage analysis can be found in Appendix C – MISO ERIS Analysis (CEII).

6. Stability Analysis

Stability analysis will be performed in phase 2 of the study.

7. Short Circuit Analysis

Short Circuit analysis will be performed in phase 2 of the study.

8. Affected System Impact Study

PJM, CE, DTE, and WPSC Affected Systems studies will be performed in phase 2 of the study.

9. Deliverability Analysis

9.1. Introduction

Generator interconnection projects have to pass Generator Deliverability Study to be granted NRIS. If the generator is deemed not fully deliverable, the customer can choose either to change the project to an Energy Resource (ER) project, reduce their NR service request amount during decision point or to proceed with the system upgrades that will make the generator fully deliverable. Generator Deliverability Study ensures that the Network Resources, on an aggregate basis, can meet the MISO aggregate load requirements during system peak condition without getting bottled up. The wind generators are tested at 100% of their maximum output level which then can be used to meet Resource Adequacy obligations, under Module E, of the MISO Transmission and Energy Market Tariff (TEMT).

MISO Generator Deliverability Study whitepaper describing the algorithm can be found in BPM 015 – Generation Interconnection, Appendix C.

9.2. Determining the MW Restriction

If one facility is overloaded based on the assessed “severe yet credible dispatch” scenario described in the study methodology, and the generator under study has a DF greater than 5%, part or all of its output is not deliverable. The restricted MW is calculated as following:

$(\text{MW restricted}) = (\text{worst loading} - \text{MW rating}) / (\text{generator sensitivity factor})$

If the result is larger than the maximum output of the generator, 100% of this generator’s output is not deliverable.

9.3. Deliverability Study Results

The limiting constraints (mon-con pairs) seen in the deliverability analysis for the 2025 Summer case are summarized in Appendix J – MISO Deliverability Analysis (CEII)

9.4. J1492

This generator is determined to be fully deliverable for 150 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 6: NRIS Results for J1492 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 6: NRIS Results for J1492

J1492 Deliverable (NRIS) Amount in 2025 Case: (Conditional on ERS and case assumptions)						0 MW(0%)							
NRIS Project#	Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mititation /Upgrade	ISD	Level of Service Attainable (MW)	Distributio n Factor	Constrain t in ERS Analysis?	Projects Associate d with ERS Constrai n	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)	MTEP Project Cost	MTEP Cost shared /project	Notes
J1492 GEN	256147 18GAINES 138 258073 CALEDON J 138 1	Raise sag to 133 SN, 133 SE		43.53	0.0568			J1492	\$1,000,000.00	\$1,000,000.00	\$0.00		
J1492 GEN	J1639 POI - Allen Junction 345kV #1&2	Build Second line from Allen - J1639. FE to upgrade limits to at least 1900 MVA (cost does not include FE costs)		57.25	0.1092			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$59,745.51	\$3,500,000.00	\$0.00		FE (First Energy costs not included) - TBD in Ph2 (AFS)
J1492 GEN	Monroe - Lallendorf 345kV	Rebuild 16.8 miles of 2-954 ACSR to 2-1431ACSR. Will require work from FE (SN 1861, SE 2157)		61.12	0.1121			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$872,024.66	\$47,600,000.00	\$0.00		
J1492 GEN	J1639 POI - Morocco 345kV	Build 8.3 miles for a second J1639 - Morocco 345kV line.		150	0.1092			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$510,104.44	\$28,000,000.00	\$0.00		

9.5. J1516

This generator is determined to be fully deliverable for 100 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 7 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 7: NRIS Results for J1516

J1516 Deliverable (NRIS) Amount in 2025 Case: (Conditional on ERIS and case assumptions)						29.02 MW(29.02%)								
NRIS Project#	Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mititation /Upgrade	ISD	Level of Service Attainable (MW)	Distributio n Factor	Constrain t in ERIS Analysis?	Projects Associate d with ERIS Constrains	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)	MTEP Project Cost	MTEP Cost shared /project	Notes	
J1516 GEN	J1639 POI - Allen Junction 345kV #1&2	Build Second line from Allen - J1639. FE to upgrade limits to at least 1900 MVA (cost does not include FE costs)		38.17	0.2358			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$86,007.27	\$3,500,000.00	\$0.00	FE (First Energy costs not included) - TBD in Ph2 (AFS)		
J1516 GEN	Monroe - Lallendorf 345kV	Rebuild 16.8 miles of 2-954 ACSR to 2-1431ACSR. Will require work from FE (SN 1861, SE 2157)		40.75	0.269			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$1,394,617.32	\$47,600,000.00	\$0.00			
J1516 GEN	J1639 POI - Morocco 345kV	Build 8.3 miles for a second J1639 - Morocco 345kV line.		63.99	0.2358			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$734,326.17	\$28,000,000.00	\$0.00			
J1516 GEN	264664 19STEPH 345 264752 19CANIF 345 1	Fixed with Caniff Phase Shifter		100	0.0631			J1516,J1525,J1526,J1557,J1557,J1664,J1665		\$0.00	\$0.00			

9.6. J1525

This generator is determined to be fully deliverable for 200 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 8 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 8: NRIS Results for J1525

J1525 Deliverable (NRIS) Amount in 2025 Case: (Conditional on ERIS and case assumptions)						58.04 MW(29.02%)								
NRIS Project#	Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mititation /Upgrade	ISD	Level of Service Attainable (MW)	Distributio n Factor	Constrain t in ERIS Analysis?	Projects Associate d with ERIS Constrains	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)	MTEP Project Cost	MTEP Cost shared /project	Notes	
J1525 GEN	J1639 POI - Allen Junction 345kV #1&2	Build Second line from Allen - J1639. FE to upgrade limits to at least 1900 MVA (cost does not include FE costs)		76.33	0.2171			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$158,409.50	\$3,500,000.00	\$0.00	FE (First Energy costs not included) - TBD in Ph2 (AFS)		
J1525 GEN	Monroe - Lallendorf 345kV	Rebuild 16.8 miles of 2-954 ACSR to 2-1431ACSR. Will require work from FE (SN 1861, SE 2157)		81.49	0.2452			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$2,542,454.77	\$47,600,000.00	\$0.00			
J1525 GEN	J1639 POI - Morocco 345kV	Build 8.3 miles for a second J1639 - Morocco 345kV line.		127.98	0.2171			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$1,352,493.02	\$28,000,000.00	\$0.00			
J1525 GEN	264664 19STEPH 345 264752 19CANIF 345 1	Fixed with Caniff Phase Shifter		200	0.0529			J1516,J1525,J1526,J1557,J1557,J1664,J1665		\$0.00	\$0.00			

9.7. J1526

This generator is determined to be fully deliverable for 200 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 9 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 9: NRIS Results for J1526

J1526 Deliverable (NRIS) Amount in 2025 Case: (Conditional on ERIS and case assumptions)							58.04 MW(29.02%)						
NRIS Project#	Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mititation /Upgrade	ISD	Level of Service Attainable (MW)	Distributio n Factor	Constrain t in ERIS Analysis?	Projects Associate d with ERIS Constrains	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)	MTEP Project Cost	MTEP Cost shared /project	Notes
J1526 GEN	J1639 POI - Allen Junction 345kV #1&2	Build Second line from Allen - J1639. FE to upgrade limits to at least 1900 MVA (cost does not include FE costs)		76.33	0.233			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$169,935.49	\$3,500,000.00	\$0.00	FE (First Energy costs not included) - TBD in Ph2 (AFS)	
J1526 GEN	Monroe - Lallendorf 345kV	Rebuild 16.8 miles of 2-954 ACSR to 2-1431ACSR. Will require work from FE (SN 1861, SE 2157)		81.49	0.2651			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$2,749,314.37	\$47,600,000.00	\$0.00		
J1526 GEN	J1639 POI - Morocco 345kV	Build 8.3 miles for a second J1639 - Morocco 345kV line.		127.98	0.233			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$1,450,901.45	\$28,000,000.00	\$0.00		
J1526 GEN	264664 19STEPH 345 264752 19CANIF 345 1	Fixed with Caniff Phase Shifter		200	0.0627			J1516,J1525,J1526,J1557,J1557,J1664,J1665		\$0.00	\$0.00		

9.8. J1527

This generator is determined to be fully deliverable for 200 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 10 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 10: NRIS Results for J1527

J1527 Deliverable (NRIS) Amount in 2025 Case: (Conditional on ERIS and case assumptions)							58.04 MW(29.02%)						
NRIS Project#	Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mititation /Upgrade	ISD	Level of Service Attainable (MW)	Distributio n Factor	Constrain t in ERIS Analysis?	Projects Associate d with ERIS Constrains	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)	MTEP Project Cost	MTEP Cost shared /project	Notes
J1527 GEN	J1639 POI - Allen Junction 345kV #1&2	Build Second line from Allen - J1639. FE to upgrade limits to at least 1900 MVA (cost does not include FE costs)		76.33	0.1106			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$80,718.45	\$3,500,000.00	\$0.00	FE (First Energy costs not included) - TBD in Ph2 (AFS)	
J1527 GEN	Monroe - Lallendorf 345kV	Rebuild 16.8 miles of 2-954 ACSR to 2-1431ACSR. Will require work from FE (SN 1861, SE 2157)		81.49	0.1143			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$1,185,165.50	\$47,600,000.00	\$0.00		
J1527 GEN	J1639 POI - Morocco 345kV	Build 8.3 miles for a second J1639 - Morocco 345kV line.		200	0.1106			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$689,170.40	\$28,000,000.00	\$0.00		

9.9. J1537

This generator is determined to be fully deliverable for 150 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 11 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 11: NRIS Results for J1537

J1537 Deliverable (NRIS) Amount in 2025 Case: (Conditional on ERS and case assumptions)							0 MW(0%)						
NRIS Project#	Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mititation /Upgrade	ISD	Level of Service Attainable (MW)	Distributio n Factor	Constrain t in ERS Analysis?	Projects Associate d with ERS Constrains	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)	MTEP Project Cost	MTEP Cost shared /project	Notes
J1537 GEN	Cornell -J1297 POI (Cornell - North Corunna Junction) 138kV	Rebuild 3.037 miles of 477 ACSR to 954 ACSR (288SN, 332 SE)		0/43.53	0.0541			J1537,J1654,J1660	\$1,365,623.95	\$5,000,000.00	\$0.00		
J1537 GEN	40893 J1297 POI 138 256115 18CORNELLB 138 1	MTEP 15798 raises limits to 264 SN, 282 SE (MTEP 19 Target A)		0/43.53	0.0506			J1537,J1654,J1660		\$0.00	\$0.00		ISD (6/1/2022)
J1537 GEN	J1639 POI - Allen Junction 345kV #1&2	Build Second line from Allen - J1639. FE to upgrade limits to at least 1900 MVA (cost does not include FE costs)		57.25	0.2007			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$109,788.76	\$3,500,000.00	\$0.00		FE (First Energy costs not included) - TBD in Ph2 (AFS)
J1537 GEN	Monroe - Lallendorf 345kV	Rebuild 16.8 miles of 2-954 ACSR to 2-1431ACSR. Will require work from FE (SN 1861, SE 2157)		61.12	0.2229			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$1,733,680.42	\$47,600,000.00	\$0.00		
J1537 GEN	J1639 POI - Morocco 345kV	Build 8.3 miles for a second J1639 - Morocco 345kV line.		133.82	0.2007			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$937,371.40	\$28,000,000.00	\$0.00		
J1537 GEN	256155 18GOSS 138 40893 J1297 POI 138 1	MTEP 15798 raises limits to 264 SN, 282 SE (MTEP 19 Target A)		150	0.0557			J1537,J1654,J1660		\$0.00	\$0.00		ISD (6/1/2022)

9.10. J1550

This generator is determined to be fully deliverable for 200 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table below shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 12: NRIS Results for J1550

J1550 Deliverable (NRIS) Amount in 2025 Case:						58.04 MW(29.02%)							
(Conditional on ERIIS and case assumptions)													
NRIS Project#	Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mititation /Upgrade	ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIIS Analysis?	Projects Associated with ERIIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)	MTEP Project Cost	MTEP Cost shared /project	Notes
J1550 GEN	J1639 POI - Allen Junction 345kV #1&2	Build Second line from Allen - J1639. FE to upgrade limits to at least 1900 MVA (cost does not include FE costs)		76.33	0.1273			J1492,J1516 ,J1525,J1526,J1527,J15	\$92,828.04	\$3,500,000.00	\$0.00		FE (First Energy costs not
J1550 GEN	Monroe - Lallendorf 345kV	Rebuild 16.8 miles of 2-954 ACSR to 2-1431ACSR. Will require work from FE (SN 1861, SE 2157)		81.49	0.1049			J1492,J1516 ,J1525,J1526,J1527,J15	\$1,087,179.38	\$47,600,000.00	\$0.00		
J1550 GEN	J1639 POI - Morocco 345kV	Build 8.3 miles for a second J1639 - Morocco 345kV line.		139.7	0.1273			J1492,J1516 ,J1525,J1526,J1527,J15	\$792,561.53	\$28,000,000.00	\$0.00		
J1550 GEN	Beecher - Knowles (Beecher - Dowling Junction) 138kV	Raise sag limit to 225 MVA SE (2 POIs)		165.08	0.1067			J1550,J1609 ,J1609,J1635,J1658	\$115,405.41	\$600,000.00	\$0.00		
J1550 GEN	J1401 POI - Wagner 138kV	Rebuild 2.11 mi of 266.8 with 954 ACSR and upgrade terminal equipment (288 SN, 332 SE)	12/31/2021	165.25	0.4454			J1550,J1609 ,J1609		\$3,900,000.00	\$0.00		this work is covered under
J1550 GEN	Beecher 500 Breaker	Increase Relay Limit to 390 MVA		180.98	0.0951			J1550,J1571 ,J1609,J1609,J1635,J16	\$7,132.41	\$100,000.00	\$0.00		
J1550 GEN	Beecher - Morocco (Beecher - MPG Junction) #2 138kV	Rebuild 12 miles of 954 aCSR to 1431 ACSR (372 SN, 432 SE)		181.57	0.0742			J1550,J1552 ,J1571,J1609,J1609,J16	\$1,663,503.40	\$19,300,000.00	\$0.00		
J1550 GEN	Beecher - Knowles (Beecher - Dowling Junction) 138kV	Riase sag limit to 257 MVA SE (7POIs)		186.2	0.1075			J1550,J1609 ,J1609,J1635,J1658	\$406,683.48	\$2,100,000.00	\$0.00		
J1550 GEN	Verona - Wagner (Wagner - Alder Creek Junction) 138kV	Rebuild 7 miles of 954 ACSR to 1431 ACSR (372 SN 432 SE)		189.92	0.4226			J1550,J1609 ,J1609,J1635,J1658	\$3,708,555.79	\$11,256,000.00	\$0.00		
J1550 GEN	Verona - Wagner (Alder Creek Junction - Barnum Creek Junction) 138kV	Rebuld 6.7 mi of 954 ACSR with 1431 ACSR and replace switch at Alder Creek (372 SN, 431 SE)		191.06	0.4226			J1550,J1609 ,J1609,J1635,J1658	\$3,558,315.79	\$10,800,000.00	\$0.00		
J1550 GEN	Beecher - Morocco (Morocco - MPG Junction) #2 138kV	Rebuild 3.5 miles of 954ACSR with 1431 ACSR (372 SN, 432 SE)		192.89	0.0742			J1550,J1552 ,J1571,J1609,J1609,J16	\$482,674.56	\$5,600,000.00	\$0.00		
J1550 GEN	Morocco Transformer #2	Install 2nd Transformer at Morocco		195.73	0.089			J1550,J1552 ,J1571,J1586,J1609,J1609,J1	\$3,176,408.53	\$39,002,014.00	\$0.00		
J1550 GEN	Coldwater - J1658 POI (Coldwater - Ash Road Junction) 138kV	Raise sag limit to 203 MVA SE (1 POI)		200	0.2403			J1550,J1609 ,J1609,J1635	\$147,121.86	\$300,000.00	\$0.00		

9.11. J1552

This generator is determined to be fully deliverable for 100 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 13 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 13: NRIS Results for J1552

J1552 Deliverable (NRIS) Amount in 2025 Case:						0 MW(0%)							
(Conditional on ERIS and case assumptions)													
NRIS Project#	Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mititation /Upgrade	ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)	MTEP Project Cost	MTEP Cost shared /project	Notes
J1552 GEN	Mud Lake - Verona #1 138kV	Rebuild (4.4 mi) Verona-MudLake #1 from 795ACSR to 1431 ACSR (372 SN, 431 SE)		0/29.02	0.3415			J1552,J1571		\$7,100,000.00	\$0.00		This NU is proposed as part of DPP2019 phase 2 ERIS mitigations. The cost is attributed to J1430, J1310, and J1248. We'd update the 954 ACSR conductor with 1431 ACSR but the cost estimate for DPP2019 wouldn't change.
J1552 GEN	Leoni - Beecher (Cement City Junciton - Seamless Junction) 138kV	A portion of this NU is covered under MTEP #15939. The remaining sag remediation of the 4/0 CUx2 has 4 POIs and is estimated at \$1,200,000.		0/29.02	0.0977			J1552	\$1,200,000.00	\$1,200,000.00	\$0.00		
J1552 GEN	Leoni - Beecher (Leoni - Seamless Junction) 138kV	Rebuild 2.7 miles of 4/0Cu and 795ACSR to 1431 ACSR (372 MVA SN, 416 MVA SE) and Terminal upgrades		0/29.02	0.113			J1552	\$5,000,000.00	\$5,000,000.00	\$0.00		
J1552 GEN	J1639 POI - Allen Junction 345kV #1&2	Build Second line from Allen - J1639. FE to upgrade limits to at least 1900 MVA (cost does not include FE costs)		38.17	0.2054			J1492,J1516,J1525, J1526,J1527,J1537, J1550,J1552,J1553, J1557,J1557,J1571, J1586,J1586,J1587, J1589,J1603,J1609, J1609,J1614,J1614, J1635,J1639,J1654, J1658,J1659,J1660, J1663,J1664,J1665, J1666,J1666,J1667	\$74,918.97	\$3,500,000.00	\$0.00		FE (First Energy costs not included) - TBD in Ph2 (AFS)
J1552 GEN	Monroe - Lallendorf 345kV	Rebuild 16.8 miles of 2-954 ACSR to 2- 1431ACSR. Will require work from FE (SN 1861, SE 2157)		40.75	0.1748			J1492,J1516,J1525, J1526,J1527,J1537, J1550,J1552,J1553, J1557,J1557,J1571, J1586,J1586,J1587, J1589,J1603,J1609, J1609,J1614,J1614, J1635,J1639,J1654, J1658,J1659,J1660, J1663,J1664,J1665, J1666,J1666,J1667	\$906,242.04	\$47,600,000.00	\$0.00		

							J1663,J1664,J1665, J1666,J1666,J1667					
J1552 GEN	J1639 POI - Morocco 345kV	Build 8.3 miles for a second J1639 - Morocco 345kV line.	49.2	0.2054			J1492,J1516,J1525, J1526,J1527,J1537, J1550,J1552,J1553, J1557,J1557,J1571, J1586,J1586,J1587, J1589,J1603,J1609, J1609,J1614,J1614, J1635,J1654,J1658, J1659,J1660,J1663, J1664,J1665,J1666, J1666,J1667	\$639,654.77	\$28,000,000.00	\$0.00		
J1552 GEN	Leoni - Beecher (Beecher - Cement City Junction) 138kV	Rebuild 0.2 miles of 477 ACSR with 954 ACSR. Raise sag of 336 ACSRx2 to 257 SE (227 SN, 257 SE) (8 POIs)	90.49	0.1048			J1552	\$2,400,000.00	\$2,400,000.00	\$0.00		
J1552 GEN	Beecher - Morocco (Beecher - MPG Junction) #2 138kV	Rebuild 12 miles of 954 aCSR to 1431 ACSR (372 SN, 432 SE)	95.53	0.0818			J1550,J1552,J1571, J1609,J1609,J1635, J1658	\$916,327.12	\$19,300,000.00	\$0.00		
J1552 GEN	Beecher - Morocco (Morocco - MPG Junction) #2 138kV	Rebuild 3.5 miles of 954ACSR with 1431 ACSR (372 SN, 432 SE)	96.44	0.0818			J1550,J1552,J1571, J1609,J1609,J1635, J1658	\$265,877.30	\$5,600,000.00	\$0.00		
J1552 GEN	Morocco Transformer #2	Install 2nd Transformer at Morocco	100	0.1241			J1550,J1552,J1571, J1586,J1586,J1609, J1609,J1635,J1658, J1659	\$2,214,563.48	\$39,002,014.00	\$0.00		

9.12. J1553

This generator is determined to be fully deliverable for 150 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 14 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 14: NRIS Results for J1553

J1553 Deliverable (NRIS) Amount in 2025 Case: (Conditional on ERIS and case assumptions)						13.62 MW(9.08%)							
NRIS Project#	Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mititation /Upgrade	ISD	Level of Service Attainable (MW)	Distributio n Factor	Constrain t in ERIS Analysis?	Projects Associate d with ERIS Constrains	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)	MTEP Project Cost	MTEP Cost shared /project	Notes
J1553 GEN	Gaines - J1292 POI (Gaines - Caledonia Junction) 138kV	Increase Sag limit to 195 MVA SE		28.24	0.0776			J1553,J1589	\$302,888.37	\$1,000,000.00	\$0.00		
J1553 GEN	Cornell - Tihart (Cornell - Bennington Junction) 138kV	Rebuild 2.33 miles of 477 ACSR to 954 ACSR and remediate sag on 795ACSR (236 SN, 236 SE) (5 POIs)		43.53	0.1215			J1553,J1589,J1654,J1660	\$2,238,232.76	\$5,700,000.00	\$0.00		
J1553 GEN	J1639 POI - Allen Junction 345kV #1&2	Build Second line from Allen - J1639. FE to upgrade limits to at least 1900 MVA (cost does not include FE costs)		57.25	0.1619			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$88,596.98	\$3,500,000.00	\$0.00	FE (First Energy costs not included) - TBD in Ph2 (AFS)	
J1553 GEN	Monroe - Lallendorf 345kV	Rebuild 16.8 miles of 2-954 ACSR to 2-1431ACSR. Will require work from FE (SN 1861, SE 2157)		60.48	0.1712			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$1,331,367.02	\$47,600,000.00	\$0.00		
J1553 GEN	256520 18SNCLKJ 138 256283 18VANATJ 138 1	MTEP 15852 raises SE to 234MVA (MTEP 19 Target A)		61.07	0.1215			J1553,J1589,J1654,J1660		\$0.00	\$0.00		ISD (12/31/2021)
J1553 GEN	256069 BENNINGTN 138 256520 18SNCLKJ 138 1	MTEP 15852 raises SE to 234MVA (MTEP 19 Target A)		61.12	0.1215			J1553,J1589,J1654,J1660		\$0.00	\$0.00		ISD (12/31/2021)
J1553 GEN	J1639 POI - Morocco 345kV	Build 8.3 miles for a second J1639 - Morocco 345kV line.		150	0.1619			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$756,436.92	\$28,000,000.00	\$0.00		

9.13. J1556
This generator is determined to be fully deliverable for 200 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis.

9.14. J1557
This generator is determined to be fully deliverable for 295 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 15 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 15: NRIS Results for J1557

J1557 Deliverable (NRIS) Amount in 2025 Case: (Conditional on ERIS and case assumptions)						85.61MW(29.02%)								
NRIS Project#	Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mititation /Upgrade	ISD	Level of Service Attainable (MW)	Distributio n Factor	Constrai n t in ERIS Analysis?	Projects Associate d with ERIS Constrai n	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)	MTEP Project Cost	MTEP Cost shared /project	Notes	
J1557 GEN	J1639 POI - Allen Junction 345kV #1&2	Build Second line from Allen - J1639. FE to upgrade limits to at least 1900 MVA (cost does not include FE costs)		56.1	0.2303			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$123,466.76	\$3,500,000.00	\$0.00	FE (First Energy costs not included) - TBD in Ph2 (AFS)		
J1557 GEN	Monroe - Lallendorf 345kV	Rebuild 16.8 miles of 2-954 ACSR to 2-1431ACSR. Will require work from FE (SN 1861, SE 2157)		59.9	0.2618			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$1,994,976.75	\$47,600,000.00	\$0.00			
J1557 GEN	J1639 POI - Morocco 345kV	Build 8.3 miles for a second J1639 - Morocco 345kV line.		94.07	0.2303			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$1,054,153.55	\$28,000,000.00	\$0.00			
J1557 GEN	264664 19STEPH 345 264752 19CANIF 345 1	Fixed with Caniff Phase Shifter		147	0.0611			J1516,J1525,J1526,J1557,J1557,J1664,J1665		\$0.00	\$0.00			
J1557 GEN1	J1639 POI - Allen Junction 345kV #1&2	Build Second line from Allen - J1639. FE to upgrade limits to at least 1900 MVA (cost does not include FE costs)		56.48	0.2303			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$124,305.68	\$3,500,000.00	\$0.00	FE (First Energy costs not included) - TBD in Ph2 (AFS)		
J1557 GEN1	Monroe - Lallendorf 345kV	Rebuild 16.8 miles of 2-954 ACSR to 2-1431ACSR. Will require work from FE (SN 1861, SE 2157)		60.3	0.2618			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$2,008,456.32	\$47,600,000.00	\$0.00			
J1557 GEN1	J1639 POI - Morocco 345kV	Build 8.3 miles for a second J1639 - Morocco 345kV line.		94.71	0.2303			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$1,061,316.19	\$28,000,000.00	\$0.00			
J1557 GEN1	264664 19STEPH 345 264752 19CANIF 345 1	Fixed with Caniff Phase Shifter		148	0.0611			J1516,J1525,J1526,J1557,J1557,J1664,J1665		\$0.00	\$0.00			

9.15. J1571

This generator is determined to be fully deliverable for 200 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 16 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 16: NRIS Results for J1571

J1571 Deliverable (NRIS) Amount in 2025 Case:						0 MW(0%)							
(Conditional on ERIS and case assumptions)													
NRIS Project#	Next Upgrade for Higher NRIS (cumulative) (i.e. All upgrades must be	Mititation /Upgrade	ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)	MTEP Project Cost	MTEP Cost shared /project	Notes
J1571 GEN	Mud Lake - Verona #1 138kV	Rebuild (4.4 mi) Verona-MudLake #1		58.04	0.0535			J1552,J1571		\$7,100,000.00	\$0.00		This NO is proposed as part of DPP2019 phase 2 ERIS mitigations. The cost is attributed to J1430,
J1571 GEN	J1639 POI - Allen Junction	Build Second line from Allen -		76.33	0.3804			J1492,J1516 ,J1525,J152 6,J1527,J15	\$277,499.30	\$3,500,000.00	\$0.00		FE (First Energy costs not included) - TBD in Ph2 (AFS)
J1571 GEN	Monroe - Lallendorf 345kV	Rebuild 16.8 miles of 2-954 ACSR to		81.49	0.2344			J1492,J1516 ,J1525,J152 6,J1527,J15	\$2,429,952.19	\$47,600,000.00	\$0.00		
J1571 GEN	J1639 POI - Morocco 345kV	Build 8.3 miles for a second		165.25	0.3804			J1492,J1516 ,J1525,J152 6,J1527,J15	\$2,369,276.28	\$28,000,000.00	\$0.00		
J1571 GEN	Beecher 500 Breaker	Increase Relay Limit to 390 MVA		180.98	0.8388			J1550,J1571 ,J1609,J160 9,J1635,J16 58	\$62,912.96	\$100,000.00	\$0.00		
J1571 GEN	Beecher - Morocco (Beecher - MPG Junction) #2 138kV	Rebuild 12 miles of 954 aCSR to 1431 ACSR (372 SN, 432 SE)		184.98	0.4673			J1550,J1552 ,J1571,J160 9,J1609,J16 35,J1658	\$10,469,429.45	\$19,300,000.00	\$0.00		
J1571 GEN	J1571 POI - Beecher 138kV	Replace terminal equipment at Beecher (288 MVA SN, 332 MVA SE)		191.06	1			J1571	\$100,000.00	\$100,000.00	\$0.00		
J1571 GEN	Beecher - Morocco (Morocco - MPG Junction) #2 138kV	Rebuild 3.5 miles of 954ACSR with 1431 ACSR (372 SN, 432 SE)		192.89	0.4673			J1550,J1552 ,J1571,J160 9,J1609,J16 35,J1658	\$3,037,761.91	\$5,600,000.00	\$0.00		
J1571 GEN	Morocco Transformer #2	Install 2nd Transformer at Morocco		200	0.474			J1550,J1552 ,J1571,J158 6,J1586,J16 09,J1609,J1 635,J1658,J 1659	\$16,917,052.19	\$39,002,014.00	\$0.00		

9.16. J1586

This generator is determined to be fully deliverable for 499 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 17 shows the NRIS results and cost estimates determined in the

NRIS analysis.

Table 17: NRIS Results for J1586

J1586 Deliverable (NRIS) Amount in 2025 Case: (Conditional on ERIS and case assumptions)						144.82 MW(29.02%)								
NRIS Project#	Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mititation /Upgrade	ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)	MTEP Project Cost	MTEP Cost shared /project	Notes	
J1586 GEN	J1639 POI - Allen Junction 345kV #1&2	Build Second line from Allen - J1639. FE to upgrade limits to at least 1900 MVA (cost does not include FE costs)		95.22	0.1598			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$145,388.04	\$3,500,000.00	\$0.00	FE (First Energy costs not included) - TBD in Ph2 (AFS)		
J1586 GEN	Monroe - Lallendorf 345kV	Rebuild 16.8 miles of 2-954 ACSR to 2-1431ACSR. Will require work from FE (SN 1861, SE 2157)		101.66	0.1529			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$1,977,868.06	\$47,600,000.00	\$0.00			
J1586 GEN	J1639 POI - Morocco 345kV	Build 8.3 miles for a second J1639 - Morocco 345kV line.		240.63	0.1598			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$1,241,316.41	\$28,000,000.00	\$0.00			
J1586 GEN	Morocco Transformer #2	Install 2nd Transformer at Morocco		249.5	0.0515			J1550,J1552,J1571,J1586,J1586,J1609,J1609,J1635,J1658,J1659	\$704,978.04	\$39,002,014.00	\$0.00			
J1586 GEN1	J1639 POI - Allen Junction 345kV #1&2	Build Second line from Allen - J1639. FE to upgrade limits to at least 1900 MVA (cost does not include FE costs)		95.22	0.1598			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$145,388.04	\$3,500,000.00	\$0.00	FE (First Energy costs not included) - TBD in Ph2 (AFS)		
J1586 GEN1	Monroe - Lallendorf 345kV	Rebuild 16.8 miles of 2-954 ACSR to 2-1431ACSR. Will require work from FE (SN 1861, SE 2157)		101.66	0.1529			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$1,977,868.06	\$47,600,000.00	\$0.00			
J1586 GEN1	J1639 POI - Morocco 345kV	Build 8.3 miles for a second J1639 - Morocco 345kV line.		240.63	0.1598			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$1,241,316.41	\$28,000,000.00	\$0.00			
J1586 GEN1	Morocco Transformer #2	Install 2nd Transformer at Morocco		249.5	0.0515			J1550,J1552,J1571,J1586,J1586,J1609,J1609,J1635,J1658,J1659	\$704,978.04	\$39,002,014.00	\$0.00			

9.17. J1587

This generator is determined to be fully deliverable for 170 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 18 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 18: NRIS Results for J1587

J1587 Deliverable (NRIS) Amount in 2025 Case: (Conditional on ERIS and case assumptions)							29.02 MW(29.02%)						
NRIS Project#	Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mititation /Upgrade	ISD	Level of Service Attainable (MW)	Distributio n Factor	Constrain t in ERIS Analysis?	Projects Associate d with ERIS Constrain	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)	MTEP Project Cost	MTEP Cost shared /project	Notes
J1587 GEN	J1639 POI - Allen Junction 345kV #1&2	Build Second line from Allen - J1639. FE to upgrade limits to at least 1900 MVA (cost does not include FE costs)		38.17	0.1117			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$40,742.21	\$3,500,000.00	\$0.00		FE (First Energy costs not included) - TBD in Ph2 (AFS)
J1587 GEN	Monroe - Lallendorf 345kV	Rebuild 16.8 miles of 2-954 ACSR to 2-1431ACSR. Will require work from FE (SN 1861, SE 2157)		40.75	0.1154			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$598,285.65	\$47,600,000.00	\$0.00		
J1587 GEN	J1639 POI - Morocco 345kV	Build 8.3 miles for a second J1639 - Morocco 345kV line.		100	0.1117			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$347,855.10	\$28,000,000.00	\$0.00		

9.18. J1589

This generator is determined to be fully deliverable for 150 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 19 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 19: NRIS Results for J1589

J1589 Deliverable (NRIS) Amount in 2025 Case: (Conditional on ERIS and case assumptions)						0 MW(0%)							
NRIS Project#	Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mititation /Upgrade	ISD	Level of Service Attainable (MW)	Distributio n Factor	Constrain t in ERIS Analysis?	Projects Associate d with ERIS Constrai n	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)	MTEP Project Cost	MTEP Cost shared /project	Notes
J1589 GEN	Gaines - J1292 POI (Gaines - Caledonia Junction) 138kV	Increase Sag limit to 195 MVA SE		28.24	0.1786			J1553,J1589	\$697,111.63	\$1,000,000.00	\$0.00		
J1589 GEN	Cornell - Tihart (Cornell - Bennington Junction) 138kV	Rebuild 2.33 miles of 477 ACSR to 954 ACSR and remediate sag on 795ACSR (236 SN, 236 SE) (5 POIs)		43.53	0.0895			J1553,J1589,J1654,J1660	\$1,649,806.03	\$5,700,000.00	\$0.00		
J1589 GEN	J1639 POI - Allen Junction 345kV #1&2	Build Second line from Allen - J1639. FE to upgrade limits to at least 1900 MVA (cost does not include FE costs)		57.25	0.1487			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$81,338.52	\$3,500,000.00	\$0.00	FE (First Energy costs not included) - TBD in Ph2 (AFS)	
J1589 GEN	Monroe - Lallendorf 345kV	Rebuild 16.8 miles of 2-954 ACSR to 2-1431ACSR. Will require work from FE (SN 1861, SE 2157)		60.48	0.1565			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$1,217,309.10	\$47,600,000.00	\$0.00		
J1589 GEN	256520 18SNCLKJ 138 256283 18VANATJ 138 1	MTEP 15852 raises SE to 234MVA (MTEP 19 Target A)		61.07	0.0895			J1553,J1589,J1654,J1660		\$0.00	\$0.00		ISD (12/31/2021)
J1589 GEN	256069 BENNINGTN 138 256520 18SNCLKJ 138 1	MTEP 15852 raises SE to 234MVA (MTEP 19 Target A)		61.12	0.0895			J1553,J1589,J1654,J1660		\$0.00	\$0.00		ISD (12/31/2021)
J1589 GEN	J1639 POI - Morocco 345kV	Build 8.3 miles for a second J1639 - Morocco 345kV line.		150	0.1487			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$694,464.53	\$28,000,000.00	\$0.00		

9.19. J1603

This generator is determined to be fully deliverable for 200 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 20 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 20: NRIS Results for J1603

J1603 Deliverable (NRIS) Amount in 2025 Case: (Conditional on ERIS and case assumptions)							58.04 MW(29.02%)						
NRIS Project#	Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mititation /Upgrade	ISD	Level of Service Attainable (MW)	Distributio n Factor	Constrain t in ERIS Analysis?	Projects Associate d with ERIS Constrains	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)	MTEP Project Cost	MTEP Cost shared /project	Notes
J1603 GEN	J1639 POI - Allen Junction 345kV #1&2	Build Second line from Allen - J1639. FE to upgrade limits to at least 1900 MVA (cost does not include FE costs)		76.33	0.1543			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$112,560.83	\$3,500,000.00	\$0.00		FE (First Energy costs not included) - TBD in Ph2 (AFS)
J1603 GEN	Monroe - Lallendorf 345kV	Rebuild 16.8 miles of 2-954 ACSR to 2-1431ACSR. Will require work from FE (SN 1861, SE 2157)		81.49	0.1674			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$1,735,754.20	\$47,600,000.00	\$0.00		
J1603 GEN	J1639 POI - Morocco 345kV	Build 8.3 miles for a second J1639 - Morocco 345kV line.		200	0.1543			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$961,039.25	\$28,000,000.00	\$0.00		

9.20. J1609

This generator is determined to be fully deliverable for 125 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 21 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 21: NRIS Results for J1609

J1609 Deliverable (NRIS) Amount in 2025 Case:							36.28 MW(29.02%)						
(Conditional on ERIS and case assumptions)													
NRIS Project#	Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mititation /Upgrade	ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)	MTEP Project Cost	MTEP Cost shared /project	Notes
J1609 GEN	J1639 POI - Allen Junction 345kV #1&2	Build Second line from Allen - J1639. FEto upgrade limits to at least 1900 MVA (cost does not include FE costs)		23.85	0.1273			J1492,J1516,J1525,J1526,J11527,J15	\$29,033.84	\$3,500,000.00	\$0.00		FE (First Energy costs not
J1609 GEN	Monroe - Lallendorf 345kV	Rebuild 16.8 miles of 2-954 ACSR to 2-1431ACSR. Will require work from FE (SN 1861, SE 2157)		25.47	0.1114			J1492,J1516,J1525,J1526,J11527,J15	\$360,837.79	\$47,600,000.00	\$0.00		
J1609 GEN	J1639 POI - Morocco 345kV	Build 8.3 miles for a second J1639 - Morocco 345kV line.		43.66	0.1273			J1492,J1516,J1525,J1526,J11527,J15	\$247,889.58	\$28,000,000.00	\$0.00		
J1609 GEN	Beecher - Knowles (Beecher - Dowling Junction) 138kV	Raise sag limit to 225 MVA SE (2 POIs)		51.59	0.0701			J1550,J1609,J1609,J1635,J11658	\$23,675.68	\$600,000.00	\$0.00		
J1609 GEN	J1401 POI - Wagner 138kV	Rebuild 2.11 mi of 266.8 with 954 ACSR and upgrade terminal equipment (288 SN, 332 SE)		51.64	0.2557			J1550,J1609,J1609		\$3,900,000.00	\$0.00		this work is covered under
J1609 GEN	Beecher 500 Breaker	Increase Relay Limit to 390 MVA		56.56	0.0624			J1550,J1571,J1609,J1609,J1635,J11658	\$1,462.48	\$100,000.00	\$0.00		
J1609 GEN	Beecher - Morocco (Beecher - MPG Junction) #2 138kV	Rebuild 12 miles of 954 aCSR to 1431 ACSR (372 SN, 432 SE)		56.74	0.0547			J1550,J1552,J1571,J1609,J1609,J1635,J11658	\$383,109.87	\$19,300,000.00	\$0.00		
J1609 GEN	Beecher - Knowles (Beecher - Dowling Junction) 138kV	Riase sag limit to 257 MVA SE (7POIs)		58.19	0.0713			J1550,J1609,J1609,J1635,J11658	\$84,363.18	\$2,100,000.00	\$0.00		
J1609 GEN	Verona - Wagner (Wagner - Alder Creek Junction) 138kV	Rebuild 7 miles of 954 ACSR to 1431 ACSR (372 SN 432 SE)		59.35	0.2405			J1550,J1609,J1609,J1635,J11658	\$659,562.11	\$11,256,000.00	\$0.00		
J1609 GEN	Verona - Wagner (Alder Creek Junction - Barnum Creek Junction) 138kV	Rebuld 6.7 mi of 954 ACSR with 1431 ACSR and replace switch at Alder Creek (372 SN, 431 SE)		59.71	0.2405			J1550,J1609,J1609,J1635,J11658	\$632,842.11	\$10,800,000.00	\$0.00		
J1609 GEN	Beecher - Morocco (Morocco - MPG Junction) #2 138kV	Rebuild 3.5 miles of 954ACSR with 1431 ACSR (372 SN, 432 SE)		60.28	0.0547			J1550,J1552,J1571,J1609,J1609,J1635,J11658	\$111,161.41	\$5,600,000.00	\$0.00		
J1609 GEN	Morocco Transformer #2	Install 2nd Transformer at Morocco		61.17	0.073			J1550,J1571,J1586,J1609,J1609,J1635,J11658	\$250,366.03	\$39,002,014.00	\$0.00		
J1609 GEN	Coldwater - J1658 POI (Coldwater - Ash Road Junction) 138kV	Raise sag limit to 203 MVA SE (1 POI)		62.5	0.1422			J1550,J1609,J1609,J1635	\$27,219.84	\$300,000.00	\$0.00		
J1609 GEN1	J1639 POI - Allen Junction 345kV #1&2	Build Second line from Allen - J1639. FE to upgrade limits to at least 1900 MVA (cost does not include FE costs)		23.85	0.1273			J1492,J1516,J1525,J1526,J11527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1588,7,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$29,033.84	\$3,500,000.00	\$0.00	FE (First Energy costs not included) - TBD in Ph2 (AFS)	
J1609 GEN1	Monroe - Lallendorf 345kV	Rebuild 16.8 miles of 2-954 ACSR to 2-1431ACSR. Will require work from FE (SN 1861, SE 2157)		25.47	0.1114			J1492,J1516,J1525,J1526,J11527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1588,7,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$360,837.79	\$47,600,000.00	\$0.00		
J1609 GEN1	J1639 POI - Morocco 345kV	Build 8.3 miles for a second J1639 - Morocco 345kV line.		43.66	0.1273			J1492,J1516,J1525,J1526,J11527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1588,7,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1665,J1666,J1666,J1667	\$247,889.58	\$28,000,000.00	\$0.00		
J1609 GEN1	Beecher - Knowles (Beecher - Dowling Junction) 138kV	Raise sag limit to 225 MVA SE (2 POIs)		51.59	0.0701			J1550,J1609,J1609,J1635,J11658	\$23,675.68	\$600,000.00	\$0.00		

9.21. J1614

This generator is determined to be fully deliverable for 250 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 22 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 22: NRIS Results for J1614

J1614 Deliverable (NRIS) Amount in 2025 Case: (Conditional on ERIS and case assumptions)							72.56 MW(29.02%)						
NRIS Project#	Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mititation /Upgrade	ISD	Level of Service Attainable (MW)	Distributio n Factor	Constrain t in ERIS Analysis?	Projects Associate d with ERIS Constrains	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)	MTEP Project Cost	MTEP Cost shared /project	Notes
J1614 GEN	J1639 POI - Allen Junction 345kV #1&2	Build Second line from Allen - J1639. FE to upgrade limits to at least 1900 MVA (cost does not include FE costs)		47.71	0.1095			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$49,933.82	\$3,500,000.00	\$0.00		FE (First Energy costs not included) - TBD in Ph2 (AFS)
J1614 GEN	Monroe - Lallendorf 345kV	Rebuild 16.8 miles of 2-954 ACSR to 2-1431ACSR. Will require work from FE (SN 1861, SE 2157)		50.93	0.1127			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$730,489.15	\$47,600,000.00	\$0.00		
J1614 GEN	J1639 POI - Morocco 345kV	Build 8.3 miles for a second J1639 - Morocco 345kV line.		125	0.1095			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$426,332.71	\$28,000,000.00	\$0.00		
J1614 GEN1	J1639 POI - Allen Junction 345kV #1&2	Build Second line from Allen - J1639. FE to upgrade limits to at least 1900 MVA (cost does not include FE costs)		47.71	0.1095			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$49,933.82	\$3,500,000.00	\$0.00		FE (First Energy costs not included) - TBD in Ph2 (AFS)
J1614 GEN1	Monroe - Lallendorf 345kV	Rebuild 16.8 miles of 2-954 ACSR to 2-1431ACSR. Will require work from FE (SN 1861, SE 2157)		50.93	0.1127			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$730,489.15	\$47,600,000.00	\$0.00		
J1614 GEN1	J1639 POI - Morocco 345kV	Build 8.3 miles for a second J1639 - Morocco 345kV line.		125	0.1095			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$426,332.71	\$28,000,000.00	\$0.00		

9.22. J1635

This generator is determined to be fully deliverable for 100 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 23 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 23: NRIS Results for J1635

J1635 Deliverable (NRIS) Amount in 2025 Case: (Conditional on ERS and case assumptions)						29.02 MW(29.02%)							
NRIS Project#	Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mititation /Upgrade	ISD	Level of Service Attainable (MW)	Distributio n Factor	Constrain t in ERS Analysis?	Projects Associate d with ERS Constrain	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)	MTEP Project Cost	MTEP Cost shared /project	Notes
J1635 GEN	J1639 POI - Allen Junction 345kV #1&2	Build Second line from Allen - J1639. FE to upgrade limits to at least 1900 MVA (cost does not include FE costs)		38.17	0.1547			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$56,426.32	\$3,500,000.00	\$0.00		FE (First Energy costs not included) - TBD in Ph2 (AFS)
J1635 GEN	Monroe - Lallendorf 345kV	Rebuild 16.8 miles of 2-954 ACSR to 2-1431ACSR. Will require work from FE (SN 1861, SE 2157)		40.75	0.1252			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$649,093.27	\$47,600,000.00	\$0.00		
J1635 GEN	J1639 POI - Morocco 345kV	Build 8.3 miles for a second J1639 - Morocco 345kV line.		69.85	0.1547			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$481,765.30	\$28,000,000.00	\$0.00		
J1635 GEN	Beecher - Knowles (Beecher - Dowling Junction) 138kV	Raise sag limit to 225 MVA SE (2 POIs)		82.63	0.1458			J1550,J1609,J1609,J1635,J1658	\$78,810.81	\$600,000.00	\$0.00		
J1635 GEN	Beecher 500 Breaker	Increase Relay Limit to 390 MVA		90.49	0.1299			J1550,J1571,J1609,J1609,J1635,J1658	\$4,871.19	\$100,000.00	\$0.00		
J1635 GEN	Beecher - Morocco (Beecher - MPG Junction) #2 138kV	Rebuild 12 miles of 954 aCSR to 1431 ACSR (372 SN, 432 SE)		90.78	0.0976			J1550,J1552,J1571,J1609,J1609,J1635,J1658	\$1,093,319.40	\$19,300,000.00	\$0.00		
J1635 GEN	Beecher - Knowles (Beecher - Dowling Junction) 138kV	Riase sag limit to 257 MVA SE (7POIs)		93.1	0.1464			J1550,J1609,J1609,J1635,J1658	\$276,923.08	\$2,100,000.00	\$0.00		
J1635 GEN	Verona - Wagner (Wagner - Alder Creek Junction) 138kV	Rebuild 7 miles of 954 ACSR to 1431 ACSR (372 SN 432 SE)		94.96	0.5207			J1550,J1609,J1609,J1635,J1658	\$2,284,989.94	\$11,256,000.00	\$0.00		
J1635 GEN	Verona - Wagner (Alder Creek Junction - Barnum Creek Junction) 138kV	Rebuld 6.7 mi of 954 ACSR with 1431 ACSR and replace switch at Alder Creek (372 SN, 431 SE)		95.53	0.5207			J1550,J1609,J1609,J1635,J1658	\$2,192,421.05	\$10,800,000.00	\$0.00		
J1635 GEN	Beecher - Morocco (Morocco - MPG Junction) #2 138kV	Rebuild 3.5 miles of 954ACSR with 1431 ACSR (372 SN, 432 SE)		96.44	0.0976			J1550,J1552,J1571,J1609,J1609,J1635,J1658	\$317,232.57	\$5,600,000.00	\$0.00		
J1635 GEN	Morocco Transformer #2	Install 2nd Transformer at Morocco		97.87	0.114			J1550,J1552,J1571,J1586,J1586,J1609,J1609,J1635,J1658,J1659	\$625,915.08	\$39,002,014.00	\$0.00		
J1635 GEN	Coldwater - J1658 POI (Coldwater - Ash Road Junction) 138kV	Raise sag limit to 203 MVA SE (1 POI)		98.13	0.3215			J1550,J1609,J1609,J1635	\$98,438.46	\$300,000.00	\$0.00		
J1635 GEN	J1320 POI - Batavia 138kV	Rebuild 6.3 miles (366 ACSR) to 954 ACSR (288 SN, 332 SE)		100	0.5127			J1635,J1658	\$3,504,338.78	\$11,500,000.00	\$0.00		

9.23. J1639

This generator is determined to be fully deliverable for 100 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 24 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 24: NRIS Results for J1639

J1639 Deliverable (NRIS) Amount in 2025 Case: (Conditional on ERS and case assumptions)							29.02 MW(29.02%)						
NRIS Project#	Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mititation /Upgrade	ISD	Level of Service Attainable (MW)	Distributio n Factor	Constrain t in ERS Analysis?	Projects Associate d with ERS Constrains	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)	MTEP Project Cost	MTEP Cost shared /project	Notes
J1639 GEN	J1639 POI - Allen Junction 345kV #1&2	Build Second line from Allen - J1639. FE to upgrade limits to at least 1900 MVA (cost does not include FE costs)		38.17	0.6046			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$220,525.86	\$3,500,000.00	\$0.00		FE (First Energy costs not included) - TBD in Ph2 (AFS)
J1639 GEN	Monroe - Lallendorf 345kV	Rebuild 16.8 miles of 2-954 ACSR to 2-1431ACSR. Will require work from FE (SN 1861, SE 2157)		100	0.2608			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$1,352,104.82	\$47,600,000.00	\$0.00		

9.24. J1654

This generator is determined to be fully deliverable for 200 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 25 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 25: NRIS Results for J1654

J1654 Deliverable (NRIS) Amount in 2025 Case: (Conditional on ERIS and case assumptions)							0 MW(0%)						
NRIS Project#	Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mititation /Upgrade	ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)	MTEP Project Cost	MTEP Cost shared /project	Notes
J1654 GEN	Cornell - J1297 POI (Cornell - North Corunna Junction) 138kV	Rebuild 3.037 miles of 477 ACSR to 954 ACSR (288SN, 332 SE)		0/37.65	0.0864			J1537,J1654,J1660	\$2,907,837.20	\$5,000,000.00	\$0.00		
J1654 GEN	256155 18GOSS 138 40893 J1297 POI 138 1	MTEP 15798 raises limits to 264 SN, 282 SE (MTEP 19 Target A)		0/37.66	0.1055			J1537,J1654,J1660		\$0.00	\$0.00		ISD (6/1/2022)
J1654 GEN	40893 J1297 POI 138 256115 18CORNELLB 138 1	MTEP 15798 raises limits to 264 SN, 282 SE (MTEP 19 Target A)		0/37.67	0.0808			J1537,J1654,J1660		\$0.00	\$0.00		ISD (6/1/2022)
J1654 GEN	Cornell - Tihart (Cornell - Bennington Junction) 138kV	Rebuild 2.33 miles of 477 ACSR to 954 ACSR and remediate sag on 795ACSR (236 SN, 236 SE) (5 POIs)		58.04	0.059			J1553,J1589,J1654,J1660	\$1,449,568.97	\$5,700,000.00	\$0.00		
J1654 GEN	J1639 POI - Allen Junction 345kV #1&2	Build Second line from Allen - J1639. FE to upgrade limits to at least 1900 MVA (cost does not include FE costs)		76.33	0.1905			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$138,968.50	\$3,500,000.00	\$0.00		FE (First Energy costs not included) - TBD in Ph2 (AFS)
J1654 GEN	Monroe - Lallendorf 345kV	Rebuild 16.8 miles of 2-954 ACSR to 2-1431ACSR. Will require work from FE (SN 1861, SE 2157)		80.63	0.211			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$2,187,838.32	\$47,600,000.00	\$0.00		
J1654 GEN	256520 18SNCLKJ 138 256283 18VANATJ 138 1	MTEP 15852 raises SE to 234MVA (MTEP 19 Target A)		81.43	0.059			J1553,J1589,J1654,J1660		\$0.00	\$0.00		ISD (12/31/2021)
J1654 GEN	256069 BENNINGTN 138 256520 18SNCLKJ 138 1	MTEP 15852 raises SE to 234MVA (MTEP 19 Target A)		81.49	0.059			J1553,J1589,J1654,J1660		\$0.00	\$0.00		ISD (12/31/2021)
J1654 GEN	J1639 POI - Morocco 345kV	Build 8.3 miles for a second J1639 - Morocco 345kV line.		200	0.1905			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$1,186,506.66	\$28,000,000.00	\$0.00		

9.25. J1658

This generator is determined to be fully deliverable for 180 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 26 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 26: NRIS Results for J1658

J1658 Deliverable (NRIS) Amount in 2025 Case: (Conditional on ERIS and case assumptions)							52.24 MW(29.02%)						
NRIS Project#	Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mititation /Upgrade	ISD	Level of Service Attainable (MW)	Distributio n Factor	Constrain t in ERIS Analysis?	Projects Associate d with ERIS Constrains	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)	MTEP Project Cost	MTEP Cost shared /project	Notes
J1658 GEN	J1639 POI - Allen Junction 345kV #1&2	Build Second line from Allen - J1639. FE to upgrade limits to at least 1900 MVA (cost does not include FE costs)		68.7	0.2326			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$152,719.45	\$3,500,000.00	\$0.00	FE (First Energy costs not included) - TBD in Ph2 (AFS)	
J1658 GEN	Monroe - Lallendorf 345kV	Rebuild 16.8 miles of 2-954 ACSR to 2-1431ACSR. Will require work from FE (SN 1861, SE 2157)		73.34	0.1664			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$1,552,743.08	\$47,600,000.00	\$0.00		
J1658 GEN	J1639 POI - Morocco 345kV	Build 8.3 miles for a second J1639 - Morocco 345kV line.		125.73	0.2326			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$1,303,911.65	\$28,000,000.00	\$0.00		
J1658 GEN	Beecher - Knowles (Beecher - Dowling Junction) 138kV	Raise sag limit to 225 MVA SE (2 POIs)		148.73	0.3684			J1550,J1609,J1609,J1635,J1658	\$358,432.43	\$600,000.00	\$0.00		
J1658 GEN	Beecher 500 Breaker	Increase Relay Limit to 390 MVA		162.88	0.3283			J1550,J1571,J1609,J1609,J1635,J1658	\$22,158.47	\$100,000.00	\$0.00		
J1658 GEN	Beecher - Morocco (Beecher - MPG Junction) #2 138kV	Rebuild 12 miles of 954 aCSR to 1431 ACSR (372 SN, 432 SE)		163.41	0.2178			J1550,J1552,J1571,J1609,J1609,J1635,J1658	\$4,391,200.88	\$19,300,000.00	\$0.00		
J1658 GEN	Beecher - Knowles (Beecher - Dowling Junction) 138kV	Riase sag limit to 257 MVA SE (7POIs)		167.58	0.3664			J1550,J1609,J1609,J1635,J1658	\$1,247,667.09	\$2,100,000.00	\$0.00		
J1658 GEN	Verona - Wagner (Wagner - Alder Creek Junction) 138kV	Rebuild 7 miles of 954 ACSR to 1431 ACSR (372 SN 432 SE)		170.93	0.4992			J1550,J1609,J1609,J1635,J1658	\$3,943,330.06	\$11,256,000.00	\$0.00		
J1658 GEN	Verona - Wagner (Alder Creek Junction - Barnum Creek Junction) 138kV	Rebuld 6.7 mi of 954 ACSR with 1431 ACSR and replace switch at Alder Creek (372 SN, 431 SE)		171.95	0.4992			J1550,J1609,J1609,J1635,J1658	\$3,783,578.95	\$10,800,000.00	\$0.00		
J1658 GEN	Beecher - Morocco (Morocco - MPG Junction) #2 138kV	Rebuild 3.5 miles of 954ACSR with 1431 ACSR (372 SN, 432 SE)		173.6	0.2178			J1550,J1552,J1571,J1609,J1609,J1635,J1658	\$1,274,130.83	\$5,600,000.00	\$0.00		
J1658 GEN	Morocco Transformer #2	Install 2nd Transformer at Morocco		176.64	0.2281			J1550,J1552,J1571,J1586,J1586,J1609,J1609,J1635,J1658,J1659	\$2,254,392.39	\$39,002,014.00	\$0.00		
J1658 GEN	J1320 POI - Batavia 138kV	Rebuild 6.3 miles (366 ACSR) to 954 ACSR (288 SN, 332 SE)		180	0.6499			J1635,J1658	\$7,995,661.22	\$11,500,000.00	\$0.00		

9.26. J1659
This generator is determined to be fully deliverable for 100 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 27 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 27: NRIS Results for J1659

J1659 Deliverable (NRIS) Amount in 2025 Case: (Conditional on ERIS and case assumptions)						29.02 MW(29.02%)								
NRIS Project#	Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mititation /Upgrade	ISD	Level of Service Attainable (MW)	Distributio n Factor	Constrain t in ERIS Analysis?	Projects Associate d with ERIS Constrains	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)	MTEP Project Cost	MTEP Cost shared /project	Notes	
J1659 GEN	J1639 POI - Allen Junction 345kV #1&2	Build Second line from Allen - J1639. FE to upgrade limits to at least 1900 MVA (cost does not include FE costs)		38.17	0.2963			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$108,074.45	\$3,500,000.00	\$0.00	FE (First Energy costs not included) - TBD in Ph2 (AFS)		
J1659 GEN	Monroe - Lallendorf 345kV	Rebuild 16.8 miles of 2-954 ACSR to 2-1431ACSR. Will require work from FE (SN 1861, SE 2157)		40.75	0.2771			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$1,436,611.37	\$47,600,000.00	\$0.00			
J1659 GEN	J1639 POI - Morocco 345kV	Build 8.3 miles for a second J1639 - Morocco 345kV line.		96.44	0.2963			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$922,734.70	\$28,000,000.00	\$0.00			
J1659 GEN	Morocco Transformer #2	Install 2nd Transformer at Morocco		100	0.0629			J1550,J1552,J1571,J1586,J1586,J1609,J1609,J1635,J1658,J1659	\$345,351.39	\$39,002,014.00	\$0.00			

9.27. J1660

This generator is determined to be fully deliverable for 50 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 28 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 28: NRIS Results for J1660

J1660 Deliverable (NRIS) Amount in 2025 Case: (Conditional on ERIS and case assumptions)							0 MW(0%)						
NRIS Project#	Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mititation /Upgrade	ISD	Level of Service Attainable (MW)	Distributio n Factor	Constrain t in ERIS Analysis?	Projects Associate d with ERIS Constrains	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)	MTEP Project Cost	MTEP Cost shared /project	Notes
J1660 GEN	Cornell - J1297 POI (Cornell - North Corunna Junction) 138kV	Rebuild 3.037 miles of 477 ACSR to 954 ACSR (288SN, 332 SE)		0/9.41	0.0864			J1537,J1654,J1660	\$726,538.85	\$5,000,000.00	\$0.00		
J1660 GEN	256155 18GOSS 138 40893 J1297 POI 138 1	MTEP 15798 raises limits to 264 SN, 282 SE (MTEP 19 Target A)		0/9.41	0.1055			J1537,J1654,J1660		\$0.00	\$0.00		ISD (6/1/2022)
J1660 GEN	40893 J1297 POI 138 256115 18CORNELLB 138 1	MTEP 15798 raises limits to 264 SN, 282 SE (MTEP 19 Target A)		0/9.41	0.0808			J1537,J1654,J1660		\$0.00	\$0.00		ISD (6/1/2022)
J1660 GEN	Cornell - Tihart (Cornell - Bennington Junction) 138kV	Rebuild 2.33 miles of 477 ACSR to 954 ACSR and remediate sag on 795ACSR (236 SN, 236 SE) (5 POIs)		14.51	0.059			J1553,J1589,J1654,J1660	\$362,392.24	\$5,700,000.00	\$0.00		
J1660 GEN	J1639 POI - Allen Junction 345kV #1&2	Build Second line from Allen - J1639. FE to upgrade limits to at least 1900 MVA (cost does not include FE costs)		19.08	0.1905			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$34,723.89	\$3,500,000.00	\$0.00		FE (First Energy costs not included) - TBD in Ph2 (AFS)
J1660 GEN	Monroe - Lallendorf 345kV	Rebuild 16.8 miles of 2-954 ACSR to 2-1431ACSR. Will require work from FE (SN 1861, SE 2157)		20.16	0.211			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$546,959.58	\$47,600,000.00	\$0.00		
J1660 GEN	256520 18SNCLKJ 138 256283 18VANATJ 138 1	MTEP 15852 raises SE to 234MVA (MTEP 19 Target A)		20.36	0.059			J1553,J1589,J1654,J1660		\$0.00	\$0.00		ISD (12/31/2021)
J1660 GEN	256069 BENNINGTN 138 256520 18SNCLKJ 138 1	MTEP 15852 raises SE to 234MVA (MTEP 19 Target A)		20.37	0.059			J1553,J1589,J1654,J1660		\$0.00	\$0.00		ISD (12/31/2021)
J1660 GEN	J1639 POI - Morocco 345kV	Build 8.3 miles for a second J1639 - Morocco 345kV line.		50	0.1905			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$296,470.95	\$28,000,000.00	\$0.00		

9.28. J1663

This generator is determined to be fully deliverable for 200 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 29 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 29: NRIS Results for J1663

J1663 Deliverable (NRIS) Amount in 2025 Case: (Conditional on ERIS and case assumptions)						58.04 MW(29.02%)								
NRIS Project#	Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mititation /Upgrade	ISD	Level of Service Attainable (MW)	Distributio n Factor	Constrain t in ERIS Analysis?	Projects Associate d with ERIS Constrains	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)	MTEP Project Cost	MTEP Cost shared /project	Notes	
J1663 GEN	J1639 POI - Allen Junction 345kV #1&2	Build Second line from Allen - J1639. FE to upgrade limits to at least 1900 MVA (cost does not include FE costs)		76.33	0.1016			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$74,116.53	\$3,500,000.00	\$0.00	FE (First Energy costs not included) - TBD in Ph2 (AFS)		
J1663 GEN	Monroe - Lallendorf 345kV	Rebuild 16.8 miles of 2-954 ACSR to 2-1431ACSR. Will require work from FE (SN 1861, SE 2157)		81.49	0.1027			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1663,J1664,J1665,J1666,J1666,J1667	\$1,064,367.79	\$47,600,000.00	\$0.00			
J1663 GEN	J1639 POI - Morocco 345kV	Build 8.3 miles for a second J1639 - Morocco 345kV line.		155.3	0.1016			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$632,803.55	\$28,000,000.00	\$0.00			
J1663 GEN	Argenta - J1586 POI 345kV	Raise sag limit to 1005 MVA SE (1 POI)		200	0.0737			J1663	\$500,000.00	\$500,000.00	\$0.00			

9.29. J1664

This generator is determined to be fully deliverable for 200 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 30 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 30: NRIS Results for J1664

J1664 Deliverable (NRIS) Amount in 2025 Case: (Conditional on ERIS and case assumptions)						58.04 MW(29.02%)								
NRIS Project#	Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mititation /Upgrade	ISD	Level of Service Attainable (MW)	Distributio n Factor	Constrain t in ERIS Analysis?	Projects Associate d with ERIS Constrains	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)	MTEP Project Cost	MTEP Cost shared /project	Notes	
J1664 GEN	J1639 POI - Allen Junction 345kV #1&2	Build Second line from Allen - J1639. FE to upgrade limits to at least 1900 MVA (cost does not include FE costs)		76.33	0.2344			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$170,956.78	\$3,500,000.00	\$0.00	FE (First Energy costs not included) - TBD in Ph2 (AFS)		
J1664 GEN	Monroe - Lallendorf 345kV	Rebuild 16.8 miles of 2-954 ACSR to 2-1431ACSR. Will require work from FE (SN 1861, SE 2157)		81.49	0.2673			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$2,771,089.06	\$47,600,000.00	\$0.00			
J1664 GEN	J1639 POI - Morocco 345kV	Build 8.3 miles for a second J1639 - Morocco 345kV line.		127.98	0.2344			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$1,459,621.18	\$28,000,000.00	\$0.00			
J1664 GEN	264664 19STEPH 345 264752 19CANIF 345 1	Fixed with Caniff Phase Shifter		200	0.0653			J1516,J1525,J1526,J1557,J1557,J1664,J1665		\$0.00	\$0.00			

9.30. J1665

This generator is determined to be fully deliverable for 50 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 31 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 31: NRIS Results for J1665

J1665 Deliverable (NRIS) Amount in 2025 Case: (Conditional on ERIS and case assumptions)						14.51 MW(29.02%)								
NRIS Project#	Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mititation /Upgrade	ISD	Level of Service Attainable (MW)	Distributio n Factor	Constrain t in ERIS Analysis?	Projects Associate d with ERIS Constrains	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)	MTEP Project Cost	MTEP Cost shared /project	Notes	
J1665 GEN	J1639 POI - Allen Junction 345kV #1&2	Build Second line from Allen - J1639. FE to upgrade limits to at least 1900 MVA (cost does not include FE costs)		19.08	0.2344			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$42,748.31	\$3,500,000.00	\$0.00	FE (First Energy costs not included) - TBD in Ph2 (AFS)		
J1665 GEN	Monroe - Lallendorf 345kV	Rebuild 16.8 miles of 2-954 ACSR to 2-1431ACSR. Will require work from FE (SN 1861, SE 2157)		20.37	0.2673			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$692,642.65	\$47,600,000.00	\$0.00			
J1665 GEN	J1639 POI - Morocco 345kV	Build 8.3 miles for a second J1639 - Morocco 345kV line.		32	0.2344			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$364,983.15	\$28,000,000.00	\$0.00			
J1665 GEN	264664 19STEPH 345 264752 19CANIF 345 1	Fixed with Caniff Phase Shifter		50	0.0653			J1516,J1525,J1526,J1557,J1557,J1664,J1665		\$0.00	\$0.00			

9.31. J1666

This generator is determined to be fully deliverable for 150 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 32 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 32: NRIS Results for J1666

J1666 Deliverable (NRIS) Amount in 2025 Case: (Conditional on ERIS and case assumptions)							87.06 MW(29.02%)						
NRIS Project#	Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mititation /Upgrade	ISD	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated with ERIS Constraint	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)	MTEP Project Cost	MTEP Cost shared /project	Notes
J1666 GEN	J1639 POI - Allen Junction 345kV #1&2	Build Second line from Allen - J1639. FE to upgrade limits to at least 1900 MVA (cost does not include FE costs)		57.25	0.2214			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$121,168.86	\$3,500,000.00	\$0.00		FE (First Energy costs not included) - TBD in Ph2 (AFS)
J1666 GEN	Monroe - Lallendorf 345kV	Rebuild 16.8 miles of 2-954 ACSR to 2-1431ACSR. Will require work from FE (SN 1861, SE 2157)		61.12	0.2189			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$1,702,573.71	\$47,600,000.00	\$0.00		
J1666 GEN	J1639 POI - Morocco 345kV	Build 8.3 miles for a second J1639 - Morocco 345kV line.		150	0.2214			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$1,034,534.15	\$28,000,000.00	\$0.00		
J1666 GEN1	J1639 POI - Allen Junction 345kV #1&2	Build Second line from Allen - J1639. FE to upgrade limits to at least 1900 MVA (cost does not include FE costs)		57.25	0.2214			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$121,168.86	\$3,500,000.00	\$0.00		FE (First Energy costs not included) - TBD in Ph2 (AFS)
J1666 GEN1	Monroe - Lallendorf 345kV	Rebuild 16.8 miles of 2-954 ACSR to 2-1431ACSR. Will require work from FE (SN 1861, SE 2157)		61.12	0.2189			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$1,702,573.71	\$47,600,000.00	\$0.00		
J1666 GEN1	J1639 POI - Morocco 345kV	Build 8.3 miles for a second J1639 - Morocco 345kV line.		150	0.2214			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$1,034,534.15	\$28,000,000.00	\$0.00		

9.32. J1667

This generator is determined to be fully deliverable for 200 MW, contingent upon the system upgrades and contingent facilities identified in the NRIS analysis. Table 33 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 33: NRIS Results for J1667

J1667 Deliverable (NRIS) Amount in 2025 Case: (Conditional on ERS and case assumptions)						0 MW(0%)							
NRIS Project#	Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Mititation /Upgrade	ISD	Level of Service Attainable (MW)	Distributio n Factor	Constrain t in ERS Analysis?	Projects Associate d with ERS Constrains	Projects Associated with NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade (\$)	MTEP Project Cost	MTEP Cost shared /project	Notes
J1667 GEN	Gallagher - Twining (Twining Withey Lake) 138kV	Covered in MTEP21 #20049 (MTEP 21 Target A)	6/1/2025	0/42.9	0.0501			J1667	\$971,000.00	\$0.00	\$971,000.00	\$971,000.00	If this MTEP project is not approved then cost will be allocated to the IC
J1667 GEN	Bullock - Shale (Bullock - Gleaner Junction) 138kV	MTEP 20018 raises limits to 288 SN, 332 SE (MTEP21 Target A)	6/1/2025	0/42.9	0.0954			J1667	\$8,701,742.26	\$0.00	\$33,000,000.00	\$8,701,742.26	If this MTEP project is not approved then cost will be allocated to the IC
J1667 GEN	Bullock - Shale (Shale - Laundra Junction) 138kV	MTEP 20018 raises limits to 288 SN, 332 SE (MTEP21 Target A)	6/1/2025	0/42.9	0.0887			J1667	\$8,099,419.25	\$0.00	\$33,000,000.00	\$8,099,419.25	If this MTEP project is not approved then cost will be allocated to the IC
J1667 GEN	Bullock - Salt River #2 138kV	MTEP 20021 Raises limit to 288 SN, 332 SE (MTEP21 Target A)	12/31/2024	58.04	0.2901			J1667	\$29,310,000.00	\$0.00	\$29,310,000.00	\$29,310,000.00	If this MTEP project is not approved then cost will be allocated to the IC
J1667 GEN	J1639 POI - Allen Junction 345kV #1&2	Build Second line from Allen - J1639. FE to upgrade limits to at least 1900 MVA (cost does not include FE costs)		76.33	0.1766			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$128,828.54	\$3,500,000.00	\$0.00	FE (First Energy costs not included) - TBD in Ph2 (AFS)	
J1667 GEN	Monroe - Lallendorf 345kV	Rebuild 16.8 miles of 2-954 ACSR to 2-1431ACSR. Will require work from FE (SN 1861, SE 2157)		78.66	0.1945			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1639,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$2,016,233.00	\$47,600,000.00	\$0.00		
J1667 GEN	Bullock - Shale (Hackett Junction - Laundra Junction) 138kV	MTEP 20018 raises limits to 288 SN, 332 SE (MTEP21 Target A)	6/1/2025	81.49	0.0887			J1667	\$8,099,419.25	\$0.00	\$33,000,000.00	\$8,099,419.25	If this MTEP project is not approved then cost will be allocated to the IC
J1667 GEN	J1639 POI - Morocco 345kV	Build 8.3 miles for a second J1639 - Morocco 345kV line.		87.09	0.1766			J1492,J1516,J1525,J1526,J1527,J1537,J1550,J1552,J1553,J1557,J1571,J1586,J1586,J1587,J1589,J1603,J1609,J1609,J1614,J1614,J1635,J1654,J1658,J1659,J1660,J1663,J1664,J1665,J1666,J1666,J1667	\$1,099,932.16	\$28,000,000.00	\$0.00		
J1667 GEN	Bullock - Shale (Gleaner Junciton - Hackett Junction) 138kV	MTEP 20018 raises limits to 288 SN, 332 SE (MTEP21 Target A)	6/1/2025	121.76	0.0887			J1667	\$8,099,419.25	\$0.00	\$33,000,000.00	\$8,099,419.25	If this MTEP project is not approved then cost will be allocated to the IC
J1667 GEN	Bullock - Salt River (Bullock - Edenville Junction) #1 138kV	MTEP 20019 raises limit to (288 SN, 332 SE) (MTEP21 Target A)	6/1/2025	200	0.3491			J1667	\$21,100,000.00	\$0.00	\$21,100,000.00	\$21,100,000.00	If this MTEP project is not approved then cost will be allocated to the IC

10. Shared Network Upgrades Analysis

Shared Network Upgrade (SNU) Analysis tests for Network Upgrades driven by higher queued interconnection projects was performed for this System Impact Study. No SNU was identified for this cycle.

11. Cost Allocation

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Network Resource Interconnection service as of the draft System Impact Study report date.

11.1. Cost Assumptions for Network Upgrades

The cost estimate for each network upgrade identified in System Impact Study was provided by the corresponding transmission owning company.

11.2. Cost Allocation Methodology

The costs of Network Upgrades (NU) for a set of generation projects (one or more sub-groups or entire group with identified NU) are allocated based on the MW impact from each project on the constrained facilities in the Study Case.

Cost Allocation Methodology for Thermal Constraints

1. With all Study Group generation projects dispatched in the Post Case, all thermal constraints are identified.
2. Distribution factor from each project on each constraint is obtained.
3. For each thermal constraint, the maximum MW contribution (increasing flow) from each project is then calculated in the Post Case without any network upgrades.
4. For each thermal constraint, the cost estimates for one or a subset of NU are provided by the corresponding Transmission Owner.
5. Then the cost of each NU is allocated based on the pro rata share of the MW contribution from each project on the constraints mitigated or partly mitigated by this NU. The methodology to determine the cost allocation of one NU is:

$$\text{Project A cost portion of NU} \\ \text{Cost of NU} = \frac{\text{Max(Proj. A MW contribution on constraint)}}{\sum_i \text{Max(Proj. i MW contrution on constraint)}}$$

6. The total NU costs for each project are calculated if more than one NU is required.

12. Revisions to the report

Dated (12/06/2021)

- J1492 – FERC 827 analysis – redone
- Updated cost details:
- J1550 & J1609:
 - ERIS cost updated as J1320-Coldwater 138 kV is attributed to J1320 (DPP2019 Ph2)
 - NRIS cost updated as J1401- Wagner 138 kV is an MTEP 11884 project whose ISD is 12/31/2021
- J1552:
 - ERIS cost updated as Rice Creek Stable rebuild & Rice Creek Vrooman Rebuild are not

- needed (constraints due to invalid contg.)
- NRIS cost updated as
 - Leoni - Beecher (Cement City Junction - Seamless Junction) 138kV cost is updated to \$1,200,000
- J1654 & J1660:
 - ERIS costs are same due to Gauss Xmer but duplicate rows removed for same element
- J1667:
 - NRIS costs updated as costs due to MTEP 21 costs shown as an FYI but not included in costs for M3.
- Tables in report and Appendices updated to reflect updated cost of J1550, J1552, J1571, J1609, J1654, J1660, J1667

Dated (12/13/2021)

- J1552 & J1571:
 - NRIS cost updated as Mud Lake Verona #1 upgrade is proposed as part of DPP2019 phase 2 ERIS mitigations. The cost is attributed to J1430, J1310, and J1248. This will be updated with the 954 ACSR conductor with 1431 ACSR but the cost estimate for DPP2019 wouldn't change.
- Tables in report and Appendices updated to reflect updated cost of J1552 & J1571

Appendix A – Cost Allocation Summary (CEII)

Appendix B – Network Upgrade Table Per Project (CEII)

Appendix C – MISO ERIS Analysis (CEII)

Appendix D- MISO Stability Analysis

Appendix E – ITC Short Circuit Analysis (CEII)

Appendix F – PJM Affected Systems Study Report (CEII)

Appendix G – CE Affected Systems Study Report (CEII)

Appendix H – DTE Affected Systems Study Report (CEII)

Appendix I – WPSC Affected Systems Study Report (CEII)

Appendix J – MISO Deliverability Analysis (CEII)

Appendix K– MISO A10-Conditionality Analysis (CEII)